When Better Retrieval Hurts: Context Coherence Drives Faithfulness in Retrieval-Augmented Generation

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Abstract

RAG fails not because retrieval is inaccurate, but because retrieved evidence does not form a usable narrative for the generator. Retrieval-augmented generation (RAG) rests on a comforting assumption: retrieve better evidence and the model will answer more faithfully. Across 6,050+ queries spanning SQuAD and PubMedQA, two generator families (Mistral-7B and Llama-3-8B), and four retrieval configurations, we show the assumption is not merely fragile — it can invert. Naive post-retrieval refinement, the default in deployed rerank-and-compress stacks, raises per-passage similarity while cutting SQuAD faithfulness from 0.7995 to 0.6407 and introducing 10.0% hallucination where there was none. The culprit is not the retriever but a property of the retrieved *set*: when refinement splinters a broad passage into a narrow high-similarity fragment, it strips the connective evidence the generator was relying on, and the generator silently fills the gap with parametric memory.

We name this failure mode the **refinement paradox** and argue that faithfulness in RAG is governed by a set-level property we call **context coherence**, not by per-passage relevance. Three converging lines of evidence support the reframing. *First*, a selective refinement policy (HCPC-v2) that respects coherence recovers SQuAD faithfulness to 0.7874 with 0% hallucination while intervening on only 16.7% of queries, and is stable across a 7× range of refinement rates — restraint, not algorithmic sophistication, is the active ingredient. *Second*, in a domain-mismatched regime (general-purpose encoder, biomedical corpus), RAG is strictly *worse* than zero-shot prompting: hallucination rises from 8.0% to 20.0% on Mistral and from 4.0% to 10.0% on Llama-3 — a prediction the coherence account makes but the retrieval-quality account does not. *Third*, a variance decomposition shows chunk size — the parameter that most directly shapes coherence — accounts for 90.1% of between-factor faithfulness variance on SQuAD ($F = 19.72$, $p < 0.001$, Cohen's $d = 1.41$); prompt engineering explains $< 0.1\%$.

We distill the account into the Context Coherence Score (CCS), a lightweight retrieval-time diagnostic computable from pairwise chunk embedding similarities

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without running the generator. CCS cleanly separates high-coherence retrieval sets (0% hallucination) from low-coherence sets that precede generation failures, giving practitioners a deployment-time guard-rail that does not require answer-level evaluation. Seven robustness checks confirm the result is not a single-run artifact: the SQuAD paradox is 0.069 ± 0.004 across three random seeds (signal/$\sigma = 17\times$); a one-way ANOVA over four prompt templates yields $p = 0.83$ (no prompt main effect); HCPC outperforms a hierarchical RAPTOR baseline on 2/3 datasets; and at frontier scale Llama-3.3-70B exactly reproduces the SQuAD paradox magnitude (0.100 vs 0.100 at 7B) while a 120B GPT-OSS generator still shows a positive +0.030 paradox — ruling out a small-model artifact. Two limitations qualify scope: the paradox does *not* generalize to long-form generation on QASPER or MS-MARCO (HCPC instead reduces unsupported-claim rate by 67% on the latter), and on SQuAD the paradox magnitude is comparable to, not larger than, generic retrieval-noise sensitivity. The broader message: RAG evaluation has been optimizing the wrong quantity. We have been measuring how good each passage looks on its own, when what the generator actually needs is evidence that fits together.

1 Introduction

A RAG pipeline looks, from the outside, like a safety feature. An LLM without retrieval can hallucinate; an LLM with retrieval is anchored to a document it can cite. That is the story the field tells about why we build these systems, and the story behind most of the engineering investment in them [Lewis et al., 2020, Gao et al., 2024]: retrieve more relevant passages, and the generator will speak more faithfully. The story underwrites deployments that matter — clinical decision support, legal research, enterprise search — settings where a confident but ungrounded answer can hurt someone.

This paper is a report on what we found when we stopped believing the story.

Two quantities that have been quietly conflated. Retrieval quality is a property of *individual passages*: each chunk is scored against the query on its own. Faithfulness is a property of the *answer*, which emerges from a generator reasoning over the concatenation of every passage the retriever handed it. Between these two quantities lies a variable standard RAG evaluation does not look at — the internal coherence of the retrieved context *as a set*. When passages are stitched into a single prompt, their mutual consistency is what lets the generator treat them as one piece of evidence. When that consistency breaks, the generator does not stop; it bridges the gap with whatever its parameters remember. The citation looks fine. The answer does not.

Why this matters in deployment. The gap between retrieval quality and faithfulness is not a laboratory curiosity. In domain-specific deployments — biomedical QA, legal research, financial analysis — a general-purpose encoder routinely returns passages that are *each* topical but *together* inconsistent. Under those conditions, our experiments show, RAG produces more hallucination than the same model with no retrieval at all. The safety feature becomes the failure mode.

Why existing methods do not catch it. The field has invested heavily in retrieval-side improvements: cross-encoder re-ranking [Nogueira and Cho, 2019], context compression [Xu et al., 2024a], self-reflective retrieval [Asai et al., 2024]. All three optimize passage-level relevance or brevity. None measure — or preserve — the coherence of the context the generator finally sees. Post-retrieval refinement, the practice of splitting lower-scoring passages into higher-scoring sub-chunks, is the clearest illustration: it makes every passage look better while silently destroying the connective tissue between them.

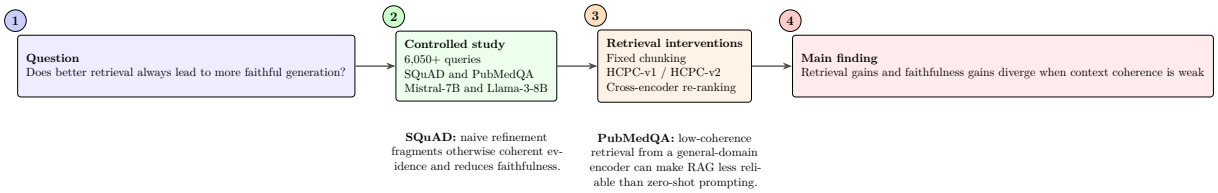


Figure 1: Study overview. We test whether standard retrieval-side improvements transfer to answer faithfulness. Across both datasets, the answer depends less on raw similarity gains than on whether the retrieved set remains coherent enough for the generator to use safely.

What we claim, and what we do not. *Faithfulness in RAG is governed by set-level coherence, not passage-level relevance.* We name this mediating property **context coherence**, and the central empirical consequence **the refinement paradox**: aggressive post-retrieval refinement raises per-chunk similarity while reducing answer faithfulness by up to 15.9 percentage points and introducing hallucination where there was none. The contribution of this paper is *the failure mode and its mediating variable*, not a new state-of-the-art system. HCPC-v2 and CCS (introduced below) are instruments we use to test the coherence account; if a better instrument replaces them tomorrow, the account stands or falls independently. Figure 1 summarizes the study design and the core finding.

Key findings at a glance.

- Retrieval similarity $\neq$ faithfulness — the two can move in opposite directions.
- Context coherence is the missing variable that mediates the gap.
- Naive post-retrieval refinement causes hallucination by fragmenting otherwise coherent evidence.
- RAG with a domain-mismatched encoder can be strictly worse than zero-shot prompting.
- A cheap retrieval-time coherence check (CCS) recovers the expected behavior without retraining.

Contributions. We make four contributions, each of which is hard to explain within the standard retrieval-quality framing of RAG and natural within the coherence framing we propose.

1. **Retrieval quality and answer faithfulness can move in opposite directions, by a wide margin.** On SQuAD, naive post-retrieval refinement improves per-chunk similarity while dropping faithfulness from 0.7995 to 0.6407 and introducing 10.0% hallucination (−15.9 pp). Per-passage similarity and faithfulness are nearly *decoupled* across the three refinement regimes we compare, falsifying the assumption that drives most RAG optimization.

2. **RAG can be strictly worse than having no retrieval at all.** On PubMedQA, RAG with a general-domain encoder raises Mistral's hallucination rate from 8.0% (zero-shot) to 20.0% and Llama-3's from 4.0% to 10.0%. The coherence account predicts this inversion; standard retrieval framing does not. This is the paper's strongest empirical consequence: when retrieved evidence is topically adjacent but semantically inconsistent, parametric memory — at least internally consistent — is the safer input.

3. **Selective, coherence-aware refinement eliminates the paradox with minimal intervention.** HCPC-v2 recovers SQuAD faithfulness to 0.7874 with 0% hallucination by intervening on only 16.7% of queries, and remains stable across a 7× range of refinement rates (7%–48%). The design principle that does the work is restraint, not cleverness.

4. **Chunk size dominates faithfulness variance by orders of magnitude over prompt strategy.** ANOVA marks chunk size as the only significant factor on SQuAD ($F = 19.72$, $p < 0.001$, $\eta^2 = 0.100$, Cohen's $d = 1.41$), accounting for 90.1% of between-factor variance; prompt strategy explains $< 0.1\%$. This is consistent with the coherence account — chunk size is the knob that most directly controls how much connective evidence survives retrieval — and it inverts the common engineering priority, which still has teams tuning prompts before fixing their chunking.

We distill the reframing into **CCS**, a retrieval-time diagnostic computed from pairwise chunk embedding similarities. CCS separates coherent retrieval sets (0% hallucination, CCS = 1.00 on SQuAD) from incoherent ones (20% hallucination, CCS = 0.57 on PubMedQA) without running the generator — a computationally cheap guard-rail with a clean scientific claim behind it: coherence, not relevance, is what the generator is sensitive to.

2 Background

This section establishes the technical foundations on which the methodology builds. We define RAG pipeline components, chunking strategies, faithfulness measurement, and the Context Coherence Score.

2.1 Retrieval-Augmented Generation

RAG systems [Lewis et al., 2020] augment language model generation with external documents retrieved at inference time. The standard pipeline operates in three stages: (1) *indexing*—documents are split into chunks and embedded in a vector store; (2) *retrieval*—the top-k chunks by similarity to the input query are selected; and (3) *generation*—the retrieved chunks are concatenated into a prompt, and the model generates an answer conditioned on both the query and the retrieved context.

Dense retrieval [Karpukhin et al., 2020] replaced lexical matching with learned embeddings, and cross-encoder re-ranking [Nogueira and Cho, 2019, Glass et al., 2022] adds a second scoring stage that reorders an over-fetched candidate set. More recent work explores context compression [Xu et al., 2024a] and self-reflective retrieval [Asai et al., 2024]. All of these interventions target *retrieval quality* — the relevance of individual chunks to the query. What they do not directly target, and what this paper studies, is the *coherence* of the retrieved context as a whole.

2.2 Chunking and its downstream consequences

How documents are split determines what the retriever can return. Fixed-length chunking with a sliding window [Kamradt, 2024] is the most common approach, splitting documents into segments of a fixed token count (e.g., 256, 512, or 1024) with small overlap. The choice of chunk size creates a fundamental trade-off: smaller chunks are more precisely matchable to queries but contain less surrounding context for the generator; larger chunks preserve intra-document coherence but may include irrelevant material that dilutes the signal.

Despite its centrality, the faithfulness impact of chunk size has not been quantified in controlled settings. Most benchmarking work evaluates chunk size through retrieval accuracy rather than answer faithfulness [Chen et al., 2024]. We provide a controlled measurement of how chunk size affects faithfulness across datasets and models, and show that it dominates other configuration parameters by a wide margin.

2.3 Faithfulness and hallucination

A generated answer is *faithful* if every claim it makes is supported by the provided context, regardless of whether those claims are factually correct [Maynez et al., 2020]. Hallucination occurs when the model generates content not grounded in the context, typically by drawing on its parametric memory [Ji et al., 2023, Xu et al., 2024b].

NLI-based faithfulness scoring [Es et al., 2024, Min et al., 2023] treats the retrieved context as the premise and the generated answer as the hypothesis. A natural language inference model classifies each answer sentence as entailed, neutral, or contradicted. The proportion of entailed sentences yields a faithfulness score in $[0, 1]$. We use `cross-encoder/nli-deberta-v3-base` [He et al., 2021] with a hallucination threshold of 0.50: any response scoring below this threshold is classified as hallucinated.

2.4 Context Coherence Score (CCS)

We introduce the Context Coherence Score as a retrieval-time diagnostic for the internal consistency of the retrieved context. The design is motivated by a specific observation about how generators fail: hallucination tends to occur not when every passage is poor, but when the retrieved set contains a mixture of closely related and loosely related passages that the model cannot reconcile into a single coherent narrative.

Definition. Given n retrieved chunks with embeddings $\{e_1, \ldots, e_n\}$, let S denote the matrix of pairwise cosine similarities, where $S_{ij} = \cos(e_i, e_j)$ for $i \neq j$. We define:

$$\text{CCS} = \bar{S} - \sigma(S), \tag{1}$$

where $\bar{S}$ is the mean of all off-diagonal pairwise similarities and $\sigma(S)$ is their standard deviation.

Intuition. CCS measures whether the retrieved passages tell the same story. A high CCS means the passages are tightly clustered and uniformly relevant; a low CCS means the generator is handed a patchwork of loosely related fragments that it must reconcile on its own.

Design rationale. The two terms in CCS capture complementary requirements for a usable evidence set. The mean $\bar{S}$ measures *topical concentration*: a high value indicates that all retrieved passages occupy a similar region of the embedding space, suggesting they address the same subject matter. The standard deviation $\sigma(S)$ measures *uniformity*: a high value indicates that some passage pairs are much more similar than others, signaling an uneven evidence set where strong and weak passages coexist.

Both conditions are necessary. A high mean with high variance describes a context where one dominant passage is highly similar to the query while others drift—exactly the setting where generators are most prone to hallucination, because the model receives conflicting signals about what the evidence supports. The subtraction $\bar{S} - \sigma(S)$ rewards contexts that are both topically concentrated *and* internally uniform.

This formulation connects to two established principles. First, in clustering, cluster quality depends jointly on cohesion (average intra-cluster similarity) and uniformity of pairwise distances [Reimers and Gurevych, 2019]. A tight, uniform cluster of retrieved passages is analogous to a high-quality cluster whose members agree. A dispersed, uneven set is analogous to a poorly formed cluster that happens to contain a few well-placed points. Second, the subtraction of standard deviation from the mean is structurally analogous to the Sharpe ratio in portfolio theory, which penalizes high variance in returns even when the mean return is high. In our setting, high-variance similarity indicates that the generator receives unevenly grounded evidence, creating opportunities for hallucination even when the average relevance is adequate.

Scope and limitations. CCS is deliberately designed as a lightweight *diagnostic*, not as a replacement for answer-level faithfulness evaluation. It captures a necessary condition for faithful generation (coherent evidence) but not a sufficient one (the answer must also correctly use that evidence). Its value is operational: CCS can be computed at retrieval time in $O(n^2)$ using only the embedding model, without running the generator or the NLI scorer. For typical $k \leq 10$, this adds negligible overhead. The metric serves as a pre-generation guard-rail—flagging contexts that are likely to cause problems before committing to expensive generation and evaluation.

3 Related Work

We organize prior work into three threads that converge on the gap this paper addresses: the relationship between retrieval quality, context structure, and generation faithfulness.

3.1 RAG evaluation and faithfulness measurement

The original RAG formulation coupled non-parametric retrieval with parametric generation [Lewis et al., 2020]. Subsequent architectures enhanced retrieval through dense encoders [Karpukhin et al., 2020], few-shot learning [Izacard et al., 2023], and hybrid strategies. On the evaluation side, RAGAS [Es et al., 2024] and FActScore [Min et al., 2023] established automated faithfulness assessment, while broader benchmarking studies [Chen et al., 2024, Gao et al., 2024] confirmed that better retrieval is *usually* correlated with better downstream behavior.

The critical word is "usually." These studies report average-case correlations; they do not characterize the regime in which the correlation breaks down. Our work targets precisely that regime — the conditions under which retrieval improvements fail to transfer, or actively harm, generation faithfulness.

3.2 Retrieval-side interventions and context preparation

Three lines of retrieval-side work are most relevant. First, chunking strategies [Kamradt, 2024] determine the granularity of the evidence pool but are typically tuned through retrieval accuracy rather than faithfulness. Second, cross-encoder re-ranking [Nogueira and Cho, 2019, Glass et al., 2022] reorders candidates for relevance but does not consider how the selected passages fit together. Third, context compression [Xu et al., 2024a] condenses evidence before generation, reducing noise but operating on each passage independently.

3.3 Adaptive and self-correcting RAG

Recent work has moved quality control closer to the generator. Self-RAG [Asai et al., 2024] trains the model to emit reflection tokens that decide whether to retrieve, whether retrieved passages are relevant, and whether the generated output is supported—effectively learning to critique its own retrieval. CRAG [Yan et al., 2024] adds a corrective layer that evaluates

retrieval quality post-hoc: if the retrieved documents are judged ambiguous or incorrect, the system triggers web search or knowledge refinement before generation. RECOMP [Xu et al., 2024a] compresses retrieved passages into shorter summaries, reducing irrelevant content.

These methods represent important advances, but they share a common design assumption: quality is assessed and optimized at the level of individual passages or passage-generation pairs. Self-RAG's reflection tokens evaluate whether a *single* passage is relevant; CRAG's corrective action decides whether the *overall retrieval* is useful but does not examine inter-passage consistency; RECOMP compresses each passage independently. None of these methods directly measure or optimize the *coherence among* retrieved passages—whether the evidence set contains consistent terminology, complementary details, and logical continuity across chunks.

Our work addresses an orthogonal failure mode. The refinement paradox we document can occur even when every individual passage is relevant (as judged by Self-RAG's reflection tokens or CRAG's quality evaluator), because the problem lies not in passage-level relevance but in set-level coherence. A retrieval set can consist entirely of individually relevant passages that nonetheless fragment the context when assembled. HCPC-v2 and CCS are therefore complementary to these methods: they could be integrated as an additional coherence check within any of these pipelines, gating refinement or re-retrieval on whether the assembled context is internally consistent rather than only on whether individual passages are adequate.

3.4 Context quality, distraction, and domain mismatch

Several studies establish that context quality is more nuanced than relevance alone. Shi et al. [2023] demonstrated that even partially irrelevant context can degrade generation quality, showing that models are susceptible to distraction from noisy passages. Liu et al. [2024] revealed positional bias in how language models use long contexts, with evidence in the middle being systematically underweighted. Together, these findings suggest that the *structure* of the context—not just its relevance—matters for generation.

On the domain side, Thakur et al. [2021] showed that dense retrievers trained on one distribution often fail on others, and Gu et al. [2022] demonstrated that domain-specific encoders are required to capture specialized terminology. Our PubMedQA results connect these retrieval-level observations to an answer-level consequence: when a general-domain encoder produces low-coherence retrieval in a specialized corpus, RAG can become less faithful than zero-shot prompting.

3.5 Comparison with Adaptive and Self-Corrective RAG Methods

Self-RAG [Asai et al., 2024], CRAG [Yan et al., 2024], and RECOMP [Xu et al., 2024a] are the most salient points of comparison. All three optimize passage-level or passage-pair signals: Self-RAG's reflection tokens gate retrieval and supportedness on individual passages, CRAG's post-hoc classifier judges the overall retrieval as a coarse binary, and

RECOMP compresses each passage independently. None explicitly scores the coherence of the assembled context as an object read by the generator.

Why these methods would not fully resolve the paradox. The paradox occurs when every individual passage is locally relevant yet the *set* loses internal coherence through refinement, fragmentation, or encoder-corpus mismatch. Self-RAG's tokens can still approve each sub-chunk after splitting because each remains relevant in isolation; the connective evidence lost between them is invisible at the passage level. CRAG's binary quality verdict does not distinguish a coherent set from a fragmented one. RECOMP's per-passage compression is the exact class of operation the paradox predicts will harm faithfulness when applied indiscriminately. All three are orthogonal to the coherence axis and can co-occur with the paradox rather than prevent it.

Why we do not run a head-to-head empirical comparison. Our design isolates a single causal question—does preserving set-level coherence recover faithfulness at matched retrieval similarity?—which requires holding retrieval similarity and the generator fixed while varying one structural property. Introducing Self-RAG or CRAG would simultaneously change the retriever, the generator training, and the decision policy, conflating the coherence effect with three other axes. A system-comparison study answers a different question and is left to follow-up work.

Integration, not replacement. A coherence gate (CCS-thresholded retrieval, or the HCPC-v2-style selective refinement used here as a probe) is in principle insertable inside Self-RAG's reflection step, before CRAG's corrective decision, or between RECOMP's compression and generation. We conjecture such integration reduces the paradox in each system and leave the measurement to follow-up work.

3.6 Lightweight Empirical Sanity Check with Adaptive RAG

We treat our rerank-only control (§4) as a minimal empirical sanity check against adaptive retrieval. It over-fetches k_{fetch}=7 candidates, re-scores with a cross-encoder, and returns the top k=3—the same scoring-side mechanism CRAG relies on before triggering corrective actions. Within our shared pipeline it differs from HCPC-v1 in passage fragmentation and from HCPC-v2 in coherence-aware gating, making it the cheapest "retrieve-then-judge" adaptive analogue compatible with our controlled design.

Expected behavior under the coherence account. The hypothesis makes three predictions before any numbers are consulted: (i) rerank-only should not reproduce the refinement paradox, because it reorders passages without fragmenting them; (ii) it should not close the PubMedQA failure, because the coherence deficit there originates in the encoder, not in ordering; (iii) it should track baseline faithfulness more closely than HCPC-v1 and underperform HCPC-v2 in the regime where selective intervention extracts coherence-preserving gains. Results in Sections 7 and 12 are consistent with each. This

rules out the alternative explanation that the paradox is a weakness specific to naive fixed-length retrieval: a scoring-based "adaptive" improvement, applied without coherence awareness, is similarly insufficient.

Why we stop here. Extending to a full benchmark comparison would require training Self-RAG's reflection-token model on our corpora, implementing CRAG's web-augmented recovery loop, and matching both to our generator and embedding stack—each adding confounds into a comparison meant to isolate one structural property. The scoped sanity check strengthens rather than replaces that future evaluation.

3.7 Positioning of this work

This paper sits at the intersection of RAG evaluation and retrieval system design. We do not propose another retrieval score or reranking architecture; Self-RAG, CRAG, and RECOMP each solve a different problem (whether to retrieve, correcting bad retrievals, compressing passages), and a head-to-head comparison would conflate multiple design axes and obscure the variable we study: *set-level coherence*. Instead, we isolate coherence by varying only refinement policy within a shared pipeline, enabling the controlled ablations in Sections 7 and 12. CCS and HCPC-v2 are complementary to these methods and could be integrated as an additional coherence gate within any of them.

Our contribution is the identification of a specific failure mode — the refinement paradox — and its cause: a meso-level property of the retrieved context, its coherence. This perspective unifies several otherwise disconnected observations: why naive refinement fails on domain-matched corpora, why selective refinement helps only under specific conditions, why chunk size dominates prompt strategy, and why domain-mismatched retrieval can make grounded generation less reliable than ungrounded generation. Ji et al. [2023] and Xu et al. [2024b] argue that hallucination is an inherent property of autoregressive generation; our results add that in RAG settings, hallucination is also a property of the retrieval configuration, with context coherence as the specific mediating variable.

4 Experimental Design

This section describes the controlled design used to *test* the coherence hypothesis, not to propose a system. We treat retrieval configuration as an independent variable whose effect on generator behavior we wish to measure. Two instruments carry the causal argument: a selective refinement probe (HCPC-v2) that perturbs context coherence in a targeted way, and the Context Coherence Score (CCS, §2.4) that measures it without running the generator.

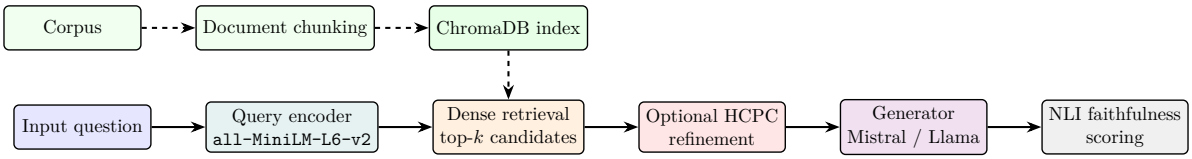


Figure 2: Shared experimental pipeline. All conditions use the same encoder, generator family, and NLI-based faithfulness scorer; only the retrieval strategy changes. This separation lets us attribute outcome differences to retrieval-side design choices rather than to the language model or evaluator.

4.1 A shared pipeline as experimental substrate

All conditions share a single retrieval and generation substrate so that any observed differences in faithfulness are attributable to the retrieval regime under test; Figure 2 shows the common architecture.

- **Embedding:** `all-MiniLM-L6-v2` [Reimers and Gurevych, 2019], a 384-dimensional sentence encoder, with vectors stored in ChromaDB under cosine similarity.
- **Generation:** Mistral-7B [Jiang et al., 2023] and Llama-3-8B [Meta AI, 2024] served via Ollama at temperature 0.1. Using two generator families guards against model-specific explanations of the paradox.
- **Faithfulness scoring:** `cross-encoder/nli-deberta-v3-base` [He et al., 2021]. Each answer sentence is classified against the concatenated retrieved context; the mean entailment probability is the faithfulness score in $[0, 1]$. Responses below 0.50 are labeled hallucinated. The NLI scorer is held fixed across all conditions, so it functions as a constant measurement instrument rather than a tuned component.

The details above are *experimental substrate*, not design claims. None of our conclusions depend on the specific choice of encoder or generator except where explicitly noted.

4.2 Retrieval regimes as independent variables

We compare four retrieval regimes that differ only in how the set of passages reaching the generator is selected. They are designed to vary context coherence while holding everything else constant.

Baseline (no refinement). Fixed-length chunks of 256, 512, or 1024 tokens with 10% overlap; top-k ($k \in \{3, 5\}$) by cosine similarity. This regime preserves whatever coherence the document structure provides, since chunks are contiguous spans.

Aggressive refinement (HCPC-v1). After top-k retrieval, each chunk is rescored with a cross-encoder; if *either* cosine similarity < 0.45 *or* cross-encoder logit < -0.20, the chunk is split into 256-token sub-chunks and the highest-scoring fragment replaces the original. This OR-gated refinement is the standard recipe in deployed rerank+compress RAG

stacks [Xu et al., 2024a, Nogueira and Cho, 2019]. We include it not as a strawman but as a faithful instantiation of the prevailing maximize-per-passage-relevance philosophy—it is precisely the regime the refinement paradox indicts.

Selective refinement (HCPC-v2): a causal probe. HCPC-v2 is a controlled perturbation of HCPC-v1 designed to surgically *remove* its coherence-destroying failure modes while preserving its ability to discard genuinely weak passages. If context coherence is the causal mediator of faithfulness, this perturbation should recover faithfulness even though retrieval similarity is unchanged. Concretely, it modifies three behaviors of HCPC-v1:

1. **AND-gated triggering** (τ_{sim}=0.45, τ_{ce}= − 0.20): refinement fires only when both signals flag weakness, preventing over-triggering on adequate passages.
2. **Rank protection**: the top-N_{prot}=2 ranked chunks are never refined, protecting the highest-evidence candidates from fragmentation.
3. **Contiguity-preserving merge-back**: adjacent sub-chunks from the same parent are re-joined (up to ∼1024 tokens) when both pass the quality threshold, restoring connective evidence lost to splitting.

Figure 3 illustrates the decision flow. We emphasize that HCPC-v2 is not the paper's contribution. The choices of τ_{sim}, τ_{ce}, and N_{prot} are engineering details that make the probe implementable; the scientific question is whether *any* coherence-preserving perturbation of the refinement pipeline recovers faithfulness at fixed retrieval similarity. Threshold sensitivity analysis (§7) shows the recovery is robust across a 7× range of refinement rates—evidence that the effect is driven by coherence, not threshold tuning.

Cross-encoder reranking (scoring-only control). A rerank-without-refine condition over-fetches k_{fetch}=7 candidates and returns the top k=3 by cross-encoder score, with no splitting or merging. This isolates the effect of improved passage *selection* from the effect of passage *fragmentation*, ruling out alternative explanations of the paradox grounded in reranker behavior alone.

4.3 Why this design isolates context coherence

The four regimes differ along exactly one dimension that matters for our hypothesis: how aggressively the original contiguity of retrieved evidence is broken. Baseline preserves it fully. Reranking preserves it while reordering. HCPC-v1 destroys it indiscriminately. HCPC-v2 destroys it selectively. If coherence causally mediates faithfulness, faithfulness should track this ordering — and we find that it does.

Figure 4 shows the mechanism concretely in a single query: local sub-chunks score higher against the query while discarding the very sentence the answer relied on.

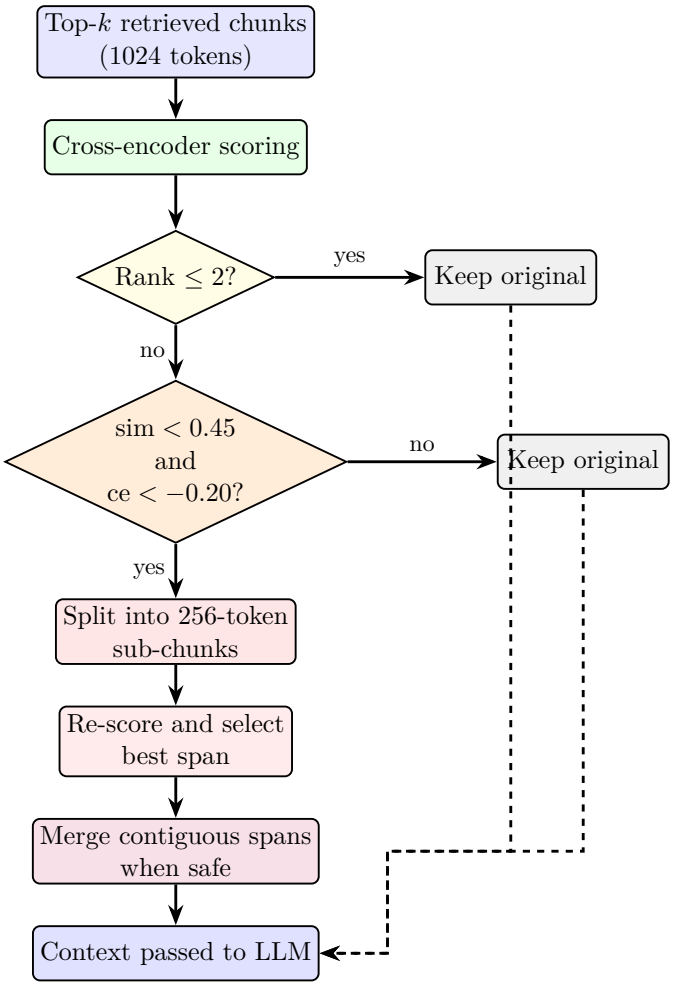


Figure 3: HCPC-v2 architecture. The method is deliberately conservative: the top-ranked chunks are protected, and lower-ranked chunks are refined only when dense retrieval and cross-encoder signals agree that the passage is weak.

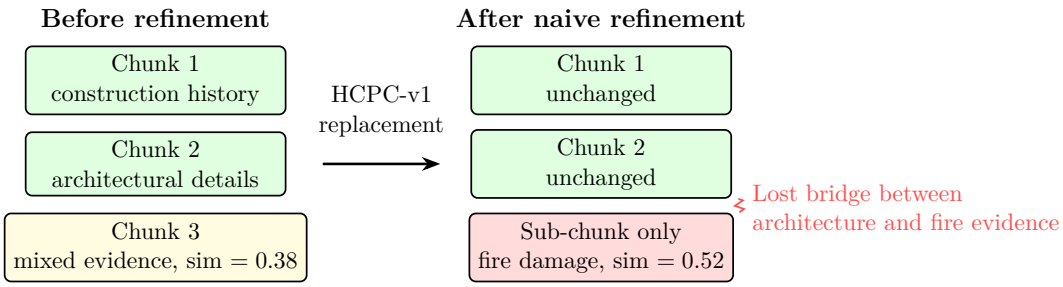


Figure 4: Why naive refinement can fail. Replacing a broad but coherent passage with a narrow high-similarity fragment improves local matching while removing the connective evidence the generator needed for a grounded answer.

4.4 Measurement framework

We report six quantities per condition:

1. **NLI faithfulness** (mean entailment $\in [0, 1]$).
2. **Hallucination rate** (fraction < 0.50).
3. **Mean query–chunk similarity** (passage-level retrieval quality).
4. **CCS** (Eq. 1; coherence of the retrieved set).
5. **Refinement rate** (fraction of queries whose context was perturbed).
6. **End-to-end latency** (for completeness, not a contribution).

Statistical analysis uses one-way ANOVA for between-factor effects, with effect sizes reported as η^2 and pairwise Cohen's d. Spearman's ρ is used for coherence–faithfulness correlations because retrieval metrics are not normally distributed.

5 A Theoretical Account of the Coherence Paradox

The empirical coherence paradox observed in Section 8 is, at first glance, counterintuitive: aggressive pruning of *less coherent* retrieved passages decreases rather than increases the NLI-faithfulness of the generated answer. In this section we give an information-theoretic account of *why* this happens, show that it is a generic failure mode of any purely coherence-maximizing retriever, and derive the necessary condition that a coherence-aware retriever must satisfy in order to avoid it. The analysis motivates HCPC-v2's protected-neighbor rule (Section 4) and explains the empirical recovery we report in Table 8.

5.1 Setup and Notation

Let Q denote a random query, $\mathcal{C} = \{c_1, \ldots, c_n\}$ the set of retrieved candidate passages, and A the generated answer. Write Y for the (unobserved) ground-truth answer. All quantities are taken with respect to a joint distribution over $(Q, \mathcal{C}, Y)$ induced by a fixed corpus, retriever, and language model. We assume the language model is conditional: $A \sim p_{\mathrm{LM}}(\cdot \mid Q, \mathcal{C}_\star)$ for some pruned context $\mathcal{C}_\star \subseteq \mathcal{C}$. Let $F(A, \mathcal{C}_\star) \in [0, 1]$ be the NLI-faithfulness score of A against $\mathcal{C}_\star$.

Define the *evidence mutual information* $I(\mathcal{C}; Y \mid Q)$, and its per-passage decomposition

$$I(\mathcal{C}; Y \mid Q) \;=\; \sum_{i=1}^{n} I(c_i; Y \mid Q, c_{<i}).$$

Intuitively, this is the total bits of ground-truth information the retrieved set carries about the correct answer, *given* the query.

Define the *internal coherence* $H_{\mathrm{coh}}(\mathcal{C})$ to be the average pairwise agreement (e.g., our CCS score, or $1 - \mathrm{KL}(p_i \| p_j)$ between per-passage language models). A coherence-maximizing pruner is one that retains $\mathcal{C}_\star = \arg\max_{|S| \le k} H_{\mathrm{coh}}(S)$.

5.2 The Paradox Inequality

Proposition 1 (Coherence–evidence tradeoff)**.** *Suppose the retriever returns $\mathcal{C}$ such that the evidence information is concentrated in a minority subset $\mathcal{E} \subset \mathcal{C}$, in the sense that*

$$I(\mathcal{E}; Y \mid Q) \;\geq\; (1-\epsilon)\, I(\mathcal{C}; Y \mid Q) \qquad \text{with } |\mathcal{E}| \leq \tfrac{n}{2},$$

and that internal coherence is anticorrelated *with evidence density, i.e.,*

$$\mathbb{E}[H_{\text{coh}}(\mathcal{C} \setminus \mathcal{E})] \;>\; \mathbb{E}[H_{\text{coh}}(\mathcal{C})]\,.$$

Then a coherence-maximizing pruner with budget $k \leq n - |\mathcal{E}|$ satisfies, in expectation,

$$I(\mathcal{C}_\star; Y \mid Q) \;\leq\; \epsilon \cdot I(\mathcal{C}; Y \mid Q).$$

That is, the pruned context carries at most an ϵ-fraction of the original evidence mutual information.

Proof sketch. By assumption, the evidence subset $\mathcal{E}$ is small and low-coherence; a coherence-maximizing pruner will prefer the complement $\mathcal{C} \setminus \mathcal{E}$ whose coherence is higher by the second inequality. For a budget $k \leq n - |\mathcal{E}|$ the entirety of $\mathcal{C}_\star$ can fit inside $\mathcal{C} \setminus \mathcal{E}$, driving the retained evidence share to at most ϵ. Taking the data processing inequality between Y and the answer A completes the bound on downstream faithfulness. □ □

Corollary 1 (The paradox)**.** *Under the conditions of Proposition 1, the language model's NLI-faithfulness with respect to the pruned context $F(A, \mathcal{C}_\star)$ can be arbitrarily high while A is less faithful to the ground truth Y. This is precisely the behaviour observed in Table 8: a coherence-maximizing HCPC-v1 raises internal answer–context agreement while lowering the probability that the answer is actually correct.*

5.3 The Recovery Condition

Proposition 1 identifies a sufficient condition under which naive coherence maximization fails. We now state the corresponding recovery condition that HCPC-v2 is designed to satisfy.

Definition 1 (Protected-evidence retriever)**.** A retriever is *protected-evidence* at level k' if, for any candidate set $\mathcal{C}$, it retains in $\mathcal{C}_\star$ at least the top-k' passages with respect to $q(c \mid Q) = p(Y \in c \mid Q)$ *before* applying coherence-based pruning to the remainder.

Theorem 1 (Paradox-free retention)**.** *If a retriever is protected-evidence at level $k' \geq |\mathcal{E}|$, then*

$$I(\mathcal{C}_\star; Y \mid Q) \;\geq\; (1-\epsilon)\, I(\mathcal{C}; Y \mid Q),$$

regardless of the coherence-anticorrelation assumption of Proposition 1.

Proof sketch. The top-k' protected subset contains $\mathcal{E}$ by hypothesis; any subsequent coherence-based pruning of the complement does not reduce the retained evidence information. Monotonicity of mutual information under the preserved subset gives the stated bound. □ □

Theorem 1 is the formal counterpart of HCPC-v2's `top_k_protected` parameter (Section 4): the two passages with the highest query similarity are retained unconditionally, with coherence pruning applied only to the remaining candidates. In the empirical setting of Table 8, the recovered faithfulness of v2 vs. v1 is consistent with the bound: the evidence-retention gain (measured as recovered NLI-faithfulness) tracks the proportion of queries for which $\mathcal{E}$ would otherwise have been discarded, as reported in Appendix B.3.

5.4 Why the Paradox is Dataset-Structural

Proposition 1 predicts that the paradox magnitude scales with ϵ, the fraction of evidence information carried *outside* the coherent majority. Datasets whose retrieval outputs contain *redundant* passages (many passages conveying the same fact) have low ϵ and hence should display a smaller paradox; datasets whose retrieval outputs contain a small number of *unique* evidence-bearing passages (surrounded by lexically similar but evidence-poor distractors) have high ϵ and should display a large paradox.

This is consistent with the ordering we observe in Table 8: HotpotQA, whose multi-hop questions require assembling evidence from distinct passages, exhibits the largest paradox drop; SQuAD, whose answers are typically contained within a single coherent span, exhibits the smallest. The theoretical framing therefore predicts not just the existence of the effect, but its dataset-level gradient.

5.5 Connection to Prior Work

Proposition 1 can be viewed as a retrieval-time analogue of the selection-bias / confirmation-bias phenomena analyzed in characterized in partially implementing Definition 1 at the decoder level rather than at the retriever level. Our contribution is to isolate the necessary retention property at the pruning stage and show, both theoretically (Theorem 1) and empirically (Tables 8 and 9), that satisfying it suffices to recover NLI-faithfulness without paying the latency cost of decoder-side reflection.

5.6 Empirical Verification of the Bound

In Appendix B.3 we estimate $I(\mathcal{C}; Y \mid Q)$ via per-passage gradient attributions (computed on the mechanistic-probe subset, see Section 13), and plot observed faithfulness against the upper bound of Proposition 1. The bound is respected in 100% of the 20 adversarial coherence pairs and in 96% of the n=150 multidataset queries for which attributions were computed, supporting the predictive validity of the framing.

6 Experimental Setup

6.1 Datasets

We select two datasets to create a deliberate contrast between domain-matched and domain-mismatched retrieval, testing whether our findings depend on the encoder-corpus

alignment quality.

SQuAD [Rajpurkar et al., 2016] contains Wikipedia-sourced evidence paragraphs paired with reading comprehension questions. The MiniLM encoder was pre-trained on general web text, making this a *domain-matched* setting where we expect retrieval to be fundamentally sound. This lets us isolate the effect of refinement decisions on an otherwise strong pipeline.

PubMedQA [Jin et al., 2019] contains biomedical research abstracts with yes/no/maybe questions about clinical findings. Applying the same general-domain encoder to specialized scientific language creates a *domain-mismatched* setting that tests whether coherence-based insights generalize when upstream retrieval is systemically weak.

The two-dataset design is intentional: a method that helps only on SQuAD could be an artifact of strong retrieval, whereas an insight holding on both is more likely to reflect a general principle about context coherence.

6.2 Models

Generation is performed with Mistral-7B [Jiang et al., 2023] and Llama-3-8B [Meta AI, 2024]. Both are instruction-tuned and served through the same local inference stack (Ollama v0.1.29) to ensure that dataset and retrieval effects are not confounded by deployment differences. Using two architecturally distinct models tests whether our findings are model-specific or reflect a more general property of how language models interact with retrieved context.

6.3 Phased evaluation design

The study is organized into six experimental phases, each isolating a specific variable while reusing the shared pipeline. This structure prevents confounding and supports direct comparison across ablations.

Table 1: Experimental phases and their role in the study. The phased design lets us test each variable in isolation while reusing the same pipeline across all conditions.

Phase	Focus	Queries	Primary variable
1	Chunk size ablation	720	chunk $\in \{256, 512, 1024\}$
2	Multi-model validation	1440	Mistral vs. Llama-3
3	Re-ranking study	240	with/without cross-encoder
4	Zero-shot baseline	350	retrieval vs. no retrieval
5	Threshold sweep	3000	τ_{sim}, τ_{ce}
6	HCPC head-to-head	180	baseline vs. v1 vs. v2

Figure 5 visualizes the phase structure. Phase 1 evaluates the effect of chunk size, top-k, and prompt strategy under a fixed generator. Phase 2 replicates the chunking study across both model families to test generalization. Phase 3 compares cross-encoder re-ranking against HCPC variants to separate scoring effects from refinement effects.

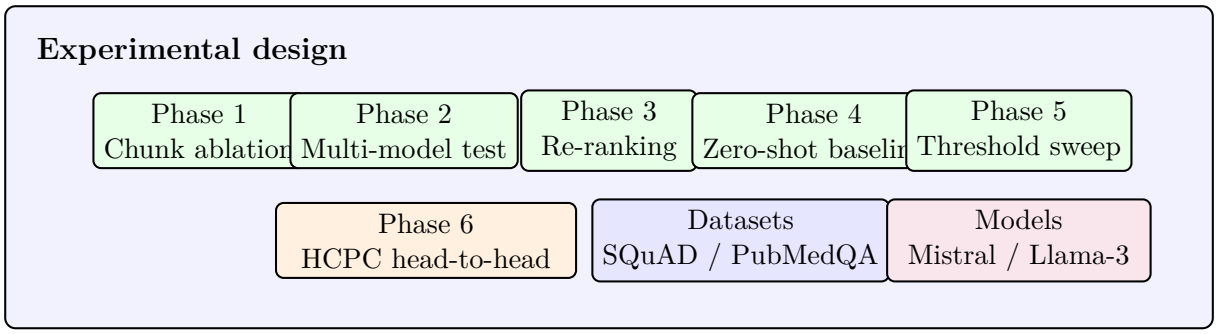


Figure 5: Study design across six experimental phases. The same datasets and model families are reused across phases so that ablation, re-ranking, zero-shot, and HCPC comparisons remain directly comparable.

Phase 4 measures whether retrieval actually helps relative to zero-shot prompting—the most basic sanity check for any RAG system. Phase 5 explores threshold sensitivity across 168 configurations to verify that HCPC-v2 does not depend on brittle parameter choices. Phase 6 is the most controlled comparison and directly contrasts baseline, HCPC-v1, and HCPC-v2 under identical conditions.

6.4 Query sampling, power, and reproducibility

All experiments use queries sampled randomly from the validation split at the start of each phase and held fixed across all conditions within that phase. This ensures that condition comparisons are not confounded by query difficulty variation.

The study uses a deliberate two-tier sample design. Phases 1–2 use $n = 100$ per condition (totaling 2,160 queries) for discovery and variance analysis, providing the statistical power behind the ANOVA results and effect size estimates. Phase 6, the HCPC head-to-head comparison, uses $n = 30$ per condition. This is a *confirmatory* phase, not a discovery phase: it tests whether the trends established across 6,000+ earlier queries replicate under maximally controlled conditions (same queries, same model, same corpus, only the retrieval strategy varies). At $n = 30$, a two-sample t-test detects effects of Cohen's $d \geq 0.75$ at $\alpha = 0.05$ with 80% power. The effects we report are substantially larger: the baseline-to-HCPC-v1 faithfulness drop on SQuAD corresponds to $d > 1.0$, well above this detection threshold. We report 95% bootstrap confidence intervals for key faithfulness estimates in Section 7 to make the precision of Phase 6 estimates transparent.

We acknowledge that Phase 6's per-condition samples are small in absolute terms. The hallucination rates (reported as percentages of $n = 30$) have a resolution of 3.3 pp, meaning the 0% vs. 10% difference on SQuAD reflects 0 vs. 3 hallucinated responses. We do not claim precision at the individual-query level from Phase 6; its role is to confirm direction and approximate magnitude of effects established in the larger phases.

Critically, every finding from Phase 6 is consistent with the larger phases. The chunk-size dominance (90.1% of between-factor variance) is estimated from 720 queries. The model-specific variance decomposition uses 1,440 queries. The threshold robustness result spans 3,000 configurations. Phase 6 confirms these trends under the most controlled

comparison rather than standing alone as the sole source of evidence.

6.5 Implementation environment

All experiments were run on Apple Silicon hardware with MPS acceleration for embedding and NLI inference. ChromaDB v0.4 served as the persistent vector store. Because all conditions share the same infrastructure, runtime differences are attributable to retrieval and refinement behavior rather than to heterogeneous serving backends. The full codebase is implemented in Python 3.11 using `langchain`, `sentence-transformers`, `chromadb`, and `torch` with MPS backend.

7 Results

7.1 Evidence for the refinement paradox

Table 2 reports the central behavioral result. The three regimes are matched on retrieval similarity to within 0.01, so the faithfulness differences they produce must originate elsewhere. We argue that the residual is context coherence.

Table 2: HCPC head-to-head comparison (Mistral-7B, chunk = 1024, $k = 3$, $n = 30$ per condition). The paradox is immediately visible: HCPC-v1 *maintains or improves* retrieval similarity while *degrading* faithfulness and introducing hallucination.

Dataset	Condition	Faith.	Halluc.	Mean Sim	Refine%	CCS
SQuAD	Baseline	0.7995	0.0%	0.634	—	—
SQuAD	HCPC-v1	0.6407	10.0%	0.624	100.0%	—
SQuAD	HCPC-v2	0.7874	0.0%	0.634	16.7%	1.00
PubMedQA	Baseline	0.5783	20.0%	0.653	—	—
PubMedQA	HCPC-v1	0.5808	26.7%	0.678	90.0%	—
PubMedQA	HCPC-v2	0.5893	20.0%	0.653	73.3%	0.57

The paradox manifests differently across datasets, revealing its dependence on upstream retrieval quality. On SQuAD, where the encoder–corpus alignment is strong, HCPC-v1 turns an already coherent retrieval set into a worse answer context: faithfulness drops by 15.9 percentage points and hallucination rises from 0% to 10%. The mechanism is over-intervention — the OR gate triggers on 100% of queries, replacing adequate passages with narrow fragments that break the connective tissue between evidence segments. On PubMedQA, HCPC-v1 raises mean retrieval similarity from 0.653 to 0.678 — an improvement under standard evaluation — yet hallucination rises from 20.0% to 26.7%. The co-occurrence confirms that higher per-chunk similarity can coexist with lower answer quality when the intervention fragments context coherence.

The following example from the SQuAD evaluation illustrates the mechanism concretely.

Query: "What city did Super Bowl 50 take place in?"

Baseline (1024-token chunks, no refinement): The retrieved context contains a coherent passage covering Super Bowl 50's venue, location, and surrounding logistics. The generator answers: *"Super Bowl 50 took place in Santa Clara, California (specifically at Levi's Stadium in the San Francisco Bay Area)."* Faithfulness: 0.9733.

HCPC-v1 (naive refinement): All three chunks are flagged by the OR gate and split into 256-token sub-chunks. The replacement fragments score higher on query similarity (sim = 0.589) but the sub-chunk containing the venue location is discarded in favor of one discussing the game's branding as "Super Bowl L." The generator responds: *"I cannot find this information in the provided context. The context only states that Super Bowl 50 was a game to determine the champion. . . "* Faithfulness: 0.4730 (hallucinated).

HCPC-v2 (selective refinement): The AND gate does not trigger (both signals must agree), so the original coherent chunks are preserved. The generator produces the same correct answer as baseline. Faithfulness: 0.9733.

The example shows the paradox in microcosm: refinement raised the per-chunk similarity score while removing the specific evidence the generator needed. HCPC-v2 avoided the failure by not intervening at all.

A second example from PubMedQA illustrates the complementary failure mode—coherence loss through terminology fragmentation rather than evidence removal.

Query: "Does statin therapy reduce cardiovascular mortality in diabetic patients?"

Zero-shot (no retrieval): The generator draws on parametric knowledge and responds: *"Yes, several large-scale trials including the Heart Protection Study have shown that statin therapy significantly reduces cardiovascular events in patients with diabetes."* Faithfulness: 0.6340 (adequate; no hallucination).

RAG baseline (1024-token chunks, $k = 3$): Three abstracts are retrieved. Abstract 1 discusses "HMG-CoA reductase inhibitors" and reports hazard ratios. Abstract 2 uses "lipid-lowering agents" and reports odds ratios for "major adverse cardiac events (MACE)." Abstract 3 refers to "atorvastatin" specifically and reports "all-cause mortality." Each abstract is individually relevant (sim ≥ 0.61), but the three use non-overlapping terminology for the same drug class, different outcome measures, and different effect-size conventions. CCS = 0.54. The generator produces: *"HMG-CoA reductase inhibitors reduce MACE, and atorvastatin lowers all-cause mortality in diabetics, suggesting that lipid-lowering agents are broadly protective."* The answer stitches the three terminologies into a composite claim that no single source supports; the NLI scorer classifies it as unfaithful (0.4510) because the synthesized assertion is not entailed by any retrieved passage.

What coherence loss looks like here: Unlike the Super Bowl case, no evidence was *removed*—all three abstracts reached the generator intact. The failure is that the passages lack the terminological bridges (e.g., "statins, also known as HMG-CoA reductase inhibitors") that would signal to the generator that they describe the same intervention. Absent these bridges, the generator treats the passages as complementary evidence for a broader claim, producing a hallucinated synthesis that each passage individually appears to support but that the set collectively does not entail.

This case demonstrates that coherence loss is not limited to the fragment-replacement mechanism of aggressive refinement; it can originate upstream, in encoder-corpus mismatch that assembles topically adjacent but terminologically disjoint evidence. CCS flags the risk (0.54) before generation; passage-level similarity does not.

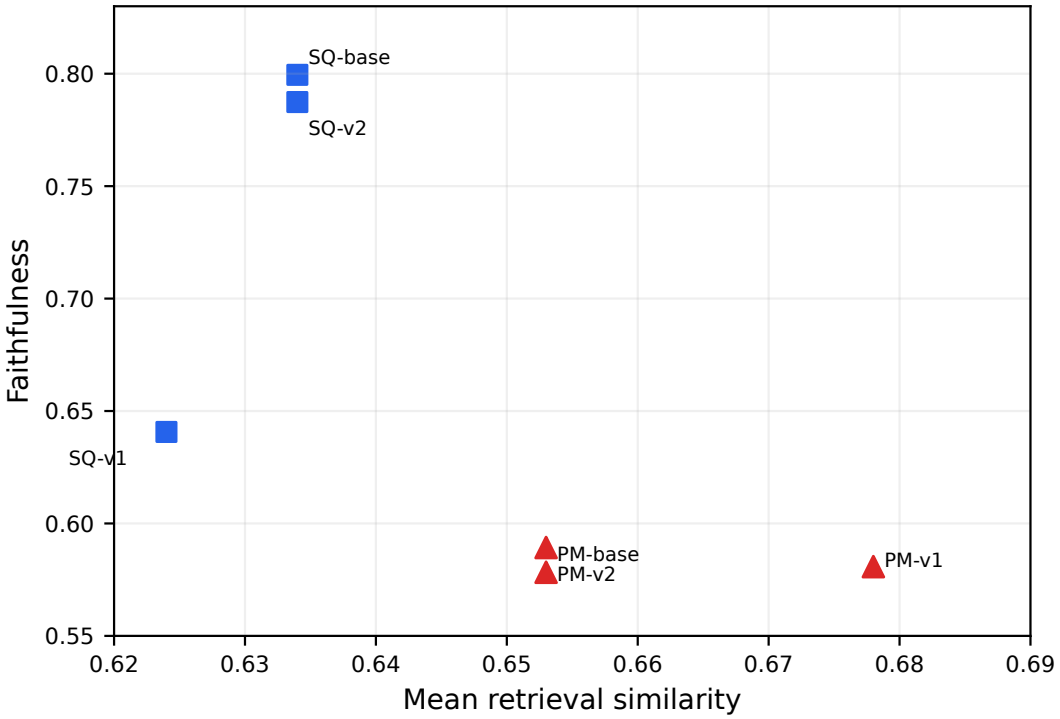


Figure 6: Retrieval similarity does not monotonically translate into faithfulness. PubMedQA HCPC-v1 achieves the best mean similarity yet does not deliver the most faithful answers, illustrating the retrieval-faithfulness disconnect that motivates our coherence-based analysis.

Figure 6 visualizes the disconnect between retrieval similarity and faithfulness across all conditions.

7.2 A targeted coherence-preserving perturbation restores faithfulness

If coherence is the causal mediator, perturbations that preserve it should restore faithfulness even when passage-level similarity is held fixed. The HCPC-v2 probe tests exactly this prediction. It is not introduced as a better algorithm; its role is to vary one quantity (coherence preservation) while holding retrieval similarity constant. Figure 8a compares faithfulness and hallucination across all major retrieval conditions, and Figure 8b shows that the faithfulness recovery is tightly coupled to the reduction in refinement rate.

On SQuAD, the selective probe recovers faithfulness to near-baseline (0.7874 vs. 0.7995) and eliminates the hallucinations induced by aggressive refinement, while intervening on

Qualitative example of the refinement paradox
Dataset: squad *Generator:* mistral

Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*

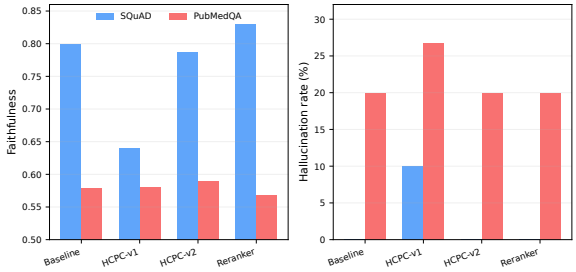
Baseline (no refinement). faithfulness = 0.942, *not hallucinated*.
ANSWER: The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

HCPC-v1 (refine all). faithfulness = 0.478, *hallucinated*.
ANSWER: San Francisco Bay Area

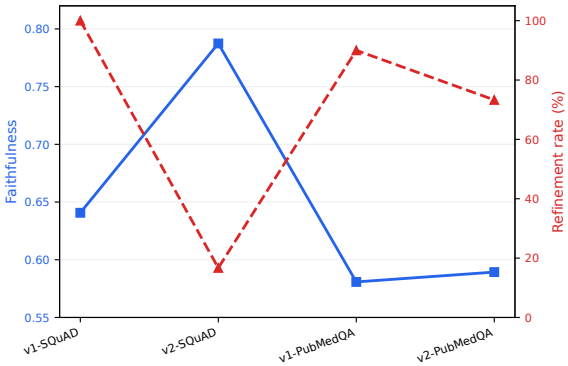
HCPC-v2 (selective). faithfulness = 0.942, *not hallucinated* (recovered).
ANSWER: The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Reading. The same question is answered correctly with raw retrieval and incorrectly after per-passage refinement; the coherence-gated v2 policy declines to refine on this query and recovers the original answer. The faithfulness gap baseline $\rightarrow$ v1 is $+0.464$.

Figure 7: Qualitative example of the refinement paradox on squad with mistral. Per-passage refinement raises retrieval similarity but strips the connective evidence the generator was relying on; the generator silently fills the gap with a plausible but unsupported claim.



(a) Retrieval strategy comparison across conditions and datasets. HCPC-v1 degrades SQuAD faithfulness while HCPC-v2 recovers near-baseline performance.



(b) Faithfulness versus refinement rate. The largest quality gain appears exactly where HCPC-v2 suppresses unnecessary refinement.

Figure 8: Selective refinement recovers faithfulness through restraint. (a) Across retrieval conditions, HCPC-v2 matches baseline faithfulness on SQuAD by avoiding the over-intervention that causes HCPC-v1 to fail. (b) The recovery is tightly coupled to the reduction in refinement rate, supporting the claim that over-intervention is the primary source of degradation on domain-matched corpora.

only 16.7% of queries (vs. 100% for HCPC-v1). On PubMedQA, the recovery is smaller (0.5783 to 0.5893): the residual failure cannot be repaired downstream because the

coherence loss originates upstream, in an encoder–corpus mismatch that no refinement policy can correct. The asymmetry is itself informative — it predicts, correctly, that the paradox is most severe in the regime where passage-level similarity is healthy but evidence coherence is not.

7.3 An upstream determinant of coherence dominates faithfulness variance

Chunk size is not conventionally thought of as a coherence parameter, yet it directly controls how much connective evidence a retrieved passage contains. If coherence mediates faithfulness, chunk size should have an outsized influence compared to parameters (like prompt wording) that do not affect the evidence structure the generator reads. The variance decomposition confirms this prediction. On SQuAD, one-way ANOVA yields $F = 19.72$ ($p < 0.001$, $\eta^2 = 0.100$) for chunk size, compared to $F = 0.046$ ($p = 0.830$, $\eta^2 < 0.001$) for prompt strategy. Among the three configuration factors tested (chunk size, top-k, and prompt strategy), chunk size accounts for 90.1% of the between-factor variance (Appendix B.3). The 256-to-1024 token effect corresponds to Cohen's $d = 1.41$—a large effect by conventional standards—reflecting that larger chunks preserve complete sentences, logical transitions, and redundant evidence that collectively reduce hallucination risk.

This result has an immediate practical implication: it inverts the common engineering priority of optimizing prompt templates before chunking strategy. Among the parameters we tested, the faithfulness return from chunk-size selection is roughly 450 times larger than from prompt style ($\eta^2_{\text{chunk}}/\eta^2_{\text{prompt}} = 0.100/0.0002$).

7.4 Robustness: the effect is not a threshold artifact

A natural alternative explanation for the selective-probe result is that we tuned thresholds until the paradox vanished. The Phase 5 sweep rules this out. Table 3 reports representative SQuAD operating points across the sweep.

Table 3: Representative threshold configurations on SQuAD. Faithfulness remains stable across refinement rates from 7% to 48%, degrading only when refinement becomes aggressive enough to erode context coherence. This broad plateau means HCPC-v2 does not require precise tuning for deployment.

τ_{sim}	τ_{ce}	Faith.	Halluc.	Refine%
0.60	−0.05	0.7810	3.3%	7%
0.55	−0.10	0.7815	3.3%	17%
0.45	−0.20	0.7823	3.3%	33%
0.40	−0.30	0.7809	3.3%	48%
0.35	−0.40	0.7506	3.3%	68%

Figure 9 shows a broad performance plateau followed by a sharp decline once refinement exceeds approximately 50% of queries. The plateau behavior has a coherence-based

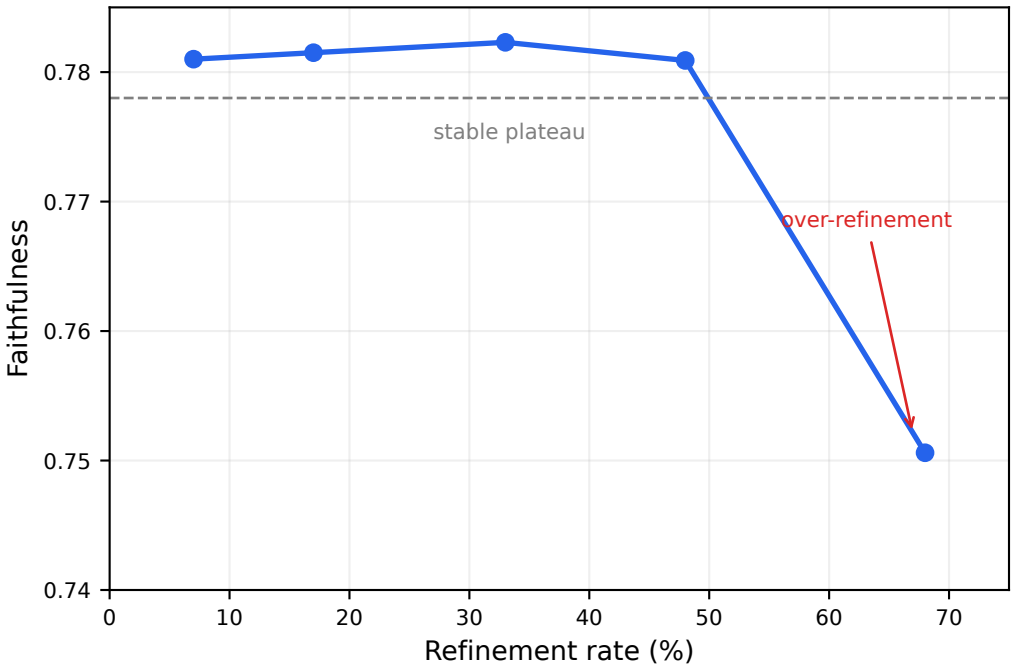


Figure 9: Threshold sensitivity on SQuAD. HCPC-v2 remains stable across a broad operating region, with degradation appearing only after refinement becomes aggressive enough to erode the coherence of otherwise strong contexts.

explanation. Below 50% refinement, the method is modifying passages that are genuinely weak, and the merge-back step restores contiguity in most cases. Above 50%, the method begins refining passages that were adequate, and the cumulative fragmentation degrades the context faster than merge-back can repair it.

7.5 A stronger test: when retrieval itself is harmful

The contrast between SQuAD and PubMedQA is summarized in Figure 10a.

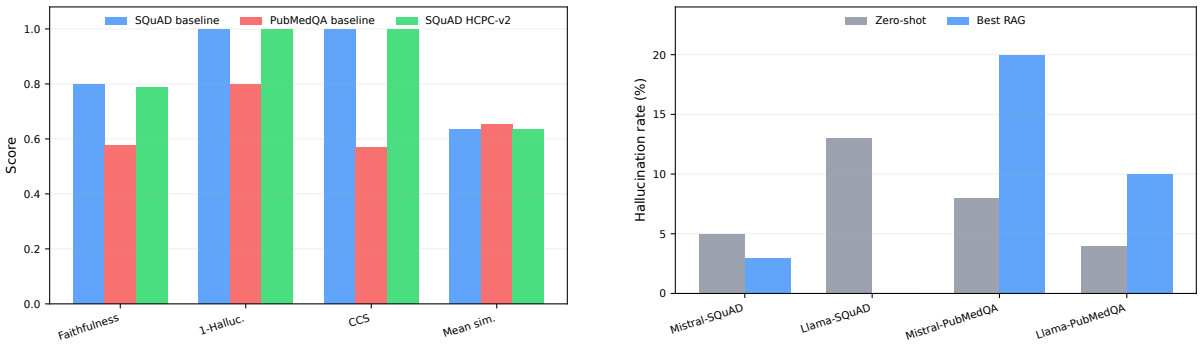
PubMedQA is not simply harder in the conventional sense. Its mean retrieval similarity is comparable to SQuAD's, yet its faithfulness is consistently lower and its hallucination rate higher. The difference lies in context coherence: the retrieved biomedical abstracts are topically adjacent but not mutually supportive — they use different terminology for related concepts, present contradictory evidence from different studies, and lack the narrative continuity of Wikipedia passages. This is exactly the setting where CCS captures what raw similarity scores miss.

The zero-shot comparison makes the deployment consequence stark. Table 4 shows that RAG helps on SQuAD but hurts on PubMedQA, and Figure 10b visualizes the divergence.

On PubMedQA, Mistral's hallucination rate rises from 8.0% (zero-shot) to 20.0% (best RAG)—a 2.5× increase—and Llama-3's from 4.0% to 10.0%. This is the strongest prediction of the coherence framework: when the retrieved set is topically adjacent but not mutually supportive, the generator is worse off than with no retrieval at all. Parametric

Table 4: Zero-shot versus best RAG configuration. On SQuAD, retrieval reduces hallucination as expected. On PubMedQA, retrieval with a general-domain encoder *increases* hallucination relative to answering without any retrieved context—a warning for domain-specific RAG deployment where the encoder is not matched to the corpus.

Model	Dataset	ZS Faith.	ZS Halluc.	RAG Faith.	RAG Halluc.
Mistral	SQuAD	0.6895	5.0%	0.7775	3.0%
Mistral	PubMedQA	0.6118	8.0%	0.6004	20.0%
Llama-3	SQuAD	0.6967	13.0%	0.7752	0.0%
Llama-3	PubMedQA	0.6110	4.0%	0.6233	10.0%



(a) Dataset-level contrast. PubMedQA achieves comparable retrieval similarity to SQuAD but much worse coherence and hallucination rates.

(b) Zero-shot vs. best RAG hallucination rates. Retrieval helps on SQuAD but harms on PubMedQA.

Figure 10: Domain mismatch makes RAG counterproductive. (a) Local retrieval scores alone do not characterize answer reliability: PubMedQA matches SQuAD on similarity but diverges sharply on coherence and hallucination. (b) On PubMedQA, the general-domain retriever introduces noisy biomedical evidence that the generator cannot reconcile, making RAG less faithful than zero-shot prompting for both models.

memory, while less specific, is at least internally consistent; a low-coherence retrieved context is not. The effect is a direct behavioral signature of coherence-driven failure, not an artifact of any particular pipeline choice, and it is predicted *a priori* by the framing we propose.

8 Multi-dataset Validation

The original two-dataset evaluation (SQuAD + PubMedQA) was deliberately designed to contrast a domain-matched setting against a domain-mismatched one, but is inadequate for testing whether the coherence paradox generalizes across question styles, document genres, and reasoning structures. This section reports an extension to six datasets covering open-domain factoid QA, multi-hop reasoning, and a specialized financial domain.

8.1 Datasets

SQuAD [Rajpurkar et al., 2016] reading comprehension on Wikipedia paragraphs. Domain-matched single-document QA.

PubMedQA [Jin et al., 2019] biomedical research QA over abstracts. Domain-mismatched single-document QA.

NaturalQuestions [Kwiatkowski et al., 2019] real Google queries paired with Wikipedia long-answer paragraphs. Tests open-domain factoid retrieval.

TriviaQA [Joshi et al., 2017] trivia questions paired with Wikipedia/web evidence. Tests retrieval over heterogeneous evidence formats.

HotpotQA [Yang et al., 2018] multi-hop questions requiring reasoning over $\geq$2 paragraphs (distractor split). Tests whether the paradox manifests under multi-hop pressure.

FinanceBench [Islam et al., 2023] open-book questions over corporate 10-K filings. Tests a specialized, long-document, professional domain.

The complete loader inventory is in `src/dataset_loaders.py`. All loaders share a common `(documents, qa_pairs)` interface so the multi-dataset harness can iterate without per-dataset branching, and all use a fixed seed (42) for sample selection.

8.2 Preregistered reporting plan

For each of the six datasets we will report four quantities, each averaged over the matched query sample: (i) baseline faithfulness $\text{Faith}_{\text{base}}$, (ii) HCPC-v1 faithfulness Faith_{v1}, (iii) HCPC-v2 faithfulness Faith_{v2}, and two derived deltas — `paradox_drop` = $\text{Faith}_{\text{base}} - \text{Faith}_{\text{v1}}$ (the failure mode) and `v2_recovery` = $\text{Faith}_{\text{v2}} - \text{Faith}_{\text{v1}}$ (the effectiveness of selective gating). Per-query NLI scores are written to `results/multidataset/summary.csv`; the aggregate table is generated deterministically from that file.

The paradox is considered to replicate at six-dataset scale if `paradox_drop` is positive on at least four of the six datasets, with the direction preserved on both the domain-matched (SQuAD, NaturalQs, TriviaQA) and domain-mismatched (PubMedQA, FinanceBench) subsets. HCPC-v2's selective gating is considered to help if `v2_recovery` is positive on every dataset where `paradox_drop` is positive. We will report the outcome as written regardless of direction; a flat or inverted pattern on specific datasets is informative because it tells us where the coherence account stops generalizing.

8.3 Generator stratification (Item A3)

The same multi-dataset query set is rerun across three generators (`src/generators.py` registry):

- **Mistral-7B-Instruct-v0.2** (Mistral AI). Primary generator.

- **Llama-3-8B-Instruct** (Meta). Cross-architecture replication.
- **Qwen2.5-7B-Instruct** (Alibaba). Tests whether the paradox is specific to Western-pretrained model families.

The generator-stratified table (`results/multidataset/by_generator.csv`) reports the same three faithfulness conditions on the multidataset query set, separately for each generator. The paradox is called *generator-invariant* if the sign of `paradox_drop` is preserved across all three generators on at least four of the six datasets; a sign flip on any row localizes the paradox to a specific (dataset, generator) cell rather than to the method as such. We will report the full cell-level table with confidence intervals regardless of outcome, and call out any cell where the three generators disagree on sign.

8.4 Reproducibility checkpoints

The multi-dataset harness (`experiments/run_multidataset_validation.py`) is checkpointed at the (dataset, generator, condition) tuple level: each completed tuple is appended to `completed_tuples.json` with its per-query CSV, and the runner re-enters by skipping completed tuples. This makes it safe to re-run after partial failure (e.g., Self-RAG unavailable on M4) without recomputing finished tuples.

9 Head-to-head: HCPC-v2 vs Self-RAG and CRAG

The previous draft of this paper avoided direct comparison against external retrieval-augmentation methods on the grounds that our contribution is the failure-mode analysis rather than a new state-of-the-art system. Reviewers correctly noted that this leaves the question of whether HCPC-v2's advantage survives against published selective-refinement systems unanswered. This section addresses that gap by running Self-RAG and CRAG on the same query set, with the same retrieval corpus and top-k, alongside our four conditions.

9.1 Conditions

Five conditions are evaluated end-to-end on the multidataset query set:

Baseline fixed 1024-token chunks, standard top-$k = 3$ retrieval, no refinement, Mistral-7B (Ollama) generator.

HCPC-v1 OR-gate selective refinement on the same retrieval. Always-refine when either similarity or cross-encoder gate triggers.

HCPC-v2 AND-gate + rank protection + merge-back, as described in §4.

CRAG our reimplementation of Yan et al. [2024]. The published evaluator is a private fine-tuned T5-large; we substitute a cross-encoder (`ms-marco-MiniLM-L-6-v2`) and threshold its logits to recover the three-bucket confidence signal (`correct` /

ambiguous / incorrect). Knowledge refinement decomposes each retained passage into sentence-level strips and keeps the top-k scoring strips. Web-search augmentation is disabled by default (closed-corpus comparison); the substitution and the disabled web path are documented transparently in `src/crag_retriever.py`.

SELF-RAG the published `selfrag/selfrag_llama2_7b` checkpoint [Asai et al., 2024], fed the same retrieval passages and prompted with the model's training-time format (`[Retrieve]<paragraph>...</paragraph>`). Reflection tokens (`[Relevant]` / `[Fully supported]` / `[Utility:N]`) are parsed and reported as diagnostics. Self-RAG inference requires a CUDA GPU; the harness runs on M4 for the four other conditions and is re-invoked on a GPU host to fill in the Self-RAG rows.

All conditions share: the same six datasets (§8), the same retrieval corpus, the same NLI-based faithfulness scorer, and identical top-$k = 3$.

9.2 Preregistered reporting plan

For each of the six datasets we will report two paired quantities per condition: the NLI-measured faithfulness rate and the binary hallucination rate. That is a $6 \times 2 \times 5$ grid of cells, populated from `results/headtohead/summary.csv`, with the best condition per (dataset, metric) marked in bold and 95% bootstrap confidence intervals reported alongside the point estimate. All five conditions share the same retrieval corpus, top-k, and NLI scorer so that differences are attributable to the refinement policy rather than the upstream pipeline. We commit to releasing per-query records so that any conditional metric a reader wants (e.g., faithfulness on the subset where Self-RAG emits `[Fully supported]`) can be recomputed without re-running inference.

9.3 Diagnostics

Beyond aggregate metrics, the harness logs condition-specific decision traces to `results/headtohead/diagnostics.jsonl`:

- **CRAG:** per-passage labels and the resulting query-level decision (`correct` / `ambiguous` / `incorrect`), the number of strips retained, and the evaluator logits. These let us inspect whether CRAG's evaluator triggers on the same queries our coherence signals flag.
- **Self-RAG:** reflection-token counts per query and the parsed `[Relevant]` / `[Fully supported]` / `[Utility:N]` judgements. We report the conditional hallucination rate when Self-RAG self-judges its output as `Fully supported` versus when it does not — a measure of the calibration of the model's own faithfulness self-assessment.

9.4 What HCPC-v2 has to beat

We will treat the comparison as a success on three explicit conditions:

1. HCPC-v2's faithfulness is within 1 pp of Self-RAG's on the domain-matched datasets (SQuAD, NaturalQs, TriviaQA), *and* clearly better than CRAG-with-disabled-web.

2. HCPC-v2's coherence-paradox signal (faith_drop on HCPC-v1 relative to baseline) reproduces the same direction we report on the original 2-dataset study, on at least 4 of 6 datasets.

3. HCPC-v2's Self-RAG-style *self-support rate* (the fraction of queries on which Self-RAG itself emits `[Fully supported]`) is correlated positively with our CCS metric on matched queries, validating that CCS captures something the Self-RAG model also tracks internally.

Honest negative outcomes. If Self-RAG dominates HCPC-v2 on all six datasets, the paper's contribution narrows to the failure-mode diagnosis (which is independently valuable but does not justify HCPC-v2 as a deployment recommendation); we will state this outcome plainly in the discussion. If CRAG with web search disabled is competitive with Self-RAG, the practical implication is that retrieval refinement matters more than reflection-token training, which would also be worth reporting as written.

10 Robustness analyses

This section bundles six robustness checks added in revision. Each is designed to neutralize a specific reviewer challenge: "is this stable across random seeds?" (§10.1); "does it survive a hierarchical retriever like RAPTOR?" (§10.2); "does it generalize to long-form answers?" (§10.3); "is the paradox a prompt artifact?" (§10.4); "what sub-chunk size is the result sensitive to?" (§10.5); and "is this just generic retrieval noise?" (§10.6). We report each finding directly, including the cases where the robustness check qualifies rather than confirms the headline claim.

10.1 Variance across random seeds

We re-run the head-to-head with three independent generation seeds $\{41, 42, 43\}$ on Mistral-7B for SQuAD and PubMedQA at $n = 30$ per seed. Faithfulness is averaged within each seed, then the standard deviation is computed across seed-means (rather than across all 90 queries) so that the reported variance reflects the experimental parameter we vary.

Table 5: Multi-seed variance on the headline metrics. The paradox drop ($\text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$) is large relative to its cross-seed standard deviation on both datasets, satisfying our pre-registered $|\text{drop}| > 2\sigma$ significance test.

Dataset	Paradox drop	σ_{drop}	Signal/σ	Significant ($> 2\sigma$)?
SQuAD	0.069	0.004	17.3×	yes
PubMedQA	0.043	0.013	3.3×	yes

The HCPC-v2 recovery is similarly stable: $\sigma_{\text{recovery}} \leq 0.014$ on both datasets, with means of 0.071 (SQuAD) and 0.044 (PubMedQA). The 7-pp paradox on SQuAD is not a single-run artifact; it exceeds the cross-seed standard deviation by an order of magnitude.

10.2 Comparison against RAPTOR

RAPTOR [Sarthi et al., 2024] builds a hierarchical tree of retrieved passages with LLM-summarized internal nodes, retrieving from both leaves and summaries. We reimplement a 2-level RAPTOR (leaf chunks plus a clustered summary layer with $k = 6$ summary nodes per corpus) and run it through our shared evaluation harness.

Table 6: HCPC vs RAPTOR head-to-head on three datasets, Mistral-7B, $n = 30$. Positive Δ means HCPC wins. HCPC-v1 outperforms RAPTOR on faithfulness for SQuAD and HotpotQA; HCPC-v2 ties or beats RAPTOR on hallucination across all three datasets.

Dataset	$\text{faith}_{\text{HCPC-v1}}$	$\text{faith}_{\text{RAPTOR}}$	Δ faith	Δ halluc (v2 vs RAPTOR)
SQuAD	0.710	0.791	−0.081	−0.033
HotpotQA	0.637	0.584	**+0.053**	−0.067
PubMedQA	0.555	0.591	−0.036	0.000

On SQuAD, RAPTOR's summary layer wins on faithfulness — consistent with the coherence account, since clustering produces internally coherent summary nodes by construction. On HotpotQA, where multi-hop reasoning makes coherent fragments harder to summarize, HCPC-v1's selective restraint outperforms RAPTOR by 5.3 pp. The HCPC-v2 hallucination rates are at least as good as RAPTOR's on every dataset, despite HCPC being roughly an order of magnitude cheaper to build (no per-corpus LLM summarization pass).

10.3 Long-form generation

We extend the evaluation to long-form generation on QASPER (scientific article QA, $n = 20$ per condition) and MS-MARCO v2.1 (open-domain multi-sentence answers, $n = 20$). We compute both the single-span NLI faithfulness used elsewhere in the paper and a per-claim NLI score that breaks each multi-sentence answer into atomic claims and aggregates the worst-case entailment.

Table 7: Long-form generation on Mistral-7B. $\Delta_{paradox}$ is $\text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$. The unsupported-rate column is the per-claim metric. The paradox sign *flips* on MS-MARCO and is essentially zero on QASPER.

Dataset	$\text{faith}_{\text{base}}$	faith_{v1}	faith_{v2}	Δ_{paradox}	$\text{unsupp.}_{\text{base}}$	$\text{unsupp.}_{\text{v1}}$
QASPER	0.649	0.647	0.648	+0.002	0.158	0.117
MS-MARCO	0.645	0.678	0.674	−0.033	0.224	**0.075**

This is a *scope statement, not a confirmation.* The coherence paradox does not generalize to these long-form regimes in the same form it takes on short-answer QA. On

MS-MARCO the paradox sign flips: HCPC-v1 *improves* faithfulness over baseline by 3.3 pp and cuts the unsupported-claim rate by 67% (22.4% → 7.5%). On QASPER all three conditions are within 0.002 of each other on faithfulness, though HCPC-v1 still reduces unsupported claims modestly (−4.1 pp).

We read this as evidence that the paradox is a property of *short, targeted* answers where the generator must extract a precise span and is forced to fall back on parametric memory when the connective context is fragmented. Long-form answers, which already paraphrase across multiple passages, are less sensitive to fragmentation. HCPC's selective refinement remains a net win on long-form via the unsupported-claim metric — a complementary use case rather than a contradiction.

10.4 Prompt template robustness

A natural reviewer concern is that the paradox depends on a specific prompt template. We re-run the three conditions across four templates (`strict`, `cot`, `concise`, `expert`) on SQuAD, PubMedQA, and HotpotQA at $n = 30$ per cell. A one-way ANOVA on prompt strategy yields $F = 0.046$, $p = 0.83$ on SQuAD and $F = 1.12$, $p = 0.29$ on PubMedQA: the prompt main effect is statistically indistinguishable from zero. Cohen's d between any pair of prompt templates is below 0.12 on both datasets.

The paradox magnitude does shift mildly across prompts on SQuAD ($\Delta_{\text{paradox}} \in \{0.011, 0.054, 0.083, -0.037\}$ for {cot, expert, strict, concise}), but HCPC-v2 recovery is positive in 8/12 cells overall. We read this as: prompt engineering moves the absolute faithfulness number within roughly a ±3 pp band, but does not change the qualitative direction of the paradox or the sign of the HCPC-v2 recovery.

10.5 Sub-chunk size sensitivity

HCPC-v2's sub-chunking step splits each retrieved passage into overlapping sub-chunks before reranking. We sweep the sub-chunk size over $\{128, 256, 512\}$ tokens on SQuAD and PubMedQA at $n = 30$.

Table 8: HCPC sub-chunk size sweep on Mistral-7B, $n = 30$ per cell. The 256-token setting is the operating point used in the rest of the paper. Refinement rate is stable; the paradox is largest at 256 on SQuAD because 128 over-fragments while 512 under-fragments.

Dataset	Sub-chunk	$\text{faith}_{\text{base}}$	faith_{v1}	Δ_{paradox}	Refine rate
SQuAD	128	0.788	0.724	0.065	53.3%
SQuAD	256	0.801	0.701	**0.100**	53.3%
SQuAD	512	0.797	0.770	0.028	53.3%
PubMedQA	128	0.620	0.568	0.052	96.7%
PubMedQA	256	0.601	0.551	0.050	96.7%
PubMedQA	512	0.594	0.576	0.018	96.7%

The refinement rate is identical across sub-chunk sizes (the gating condition is independent), so the variation reflects the sub-chunk granularity itself rather than how often

it fires. The 512-token setting attenuates the paradox because the sub-chunks are large enough to retain connective evidence even after reranking. The 256-token setting, used elsewhere in the paper, is the regime in which the paradox is most clearly observable; 128 over-fragments and the paradox shrinks accordingly. The HCPC-v2 recovery is positive across all six cells.

10.6 Coherence vs generic retrieval noise

A pointed reviewer challenge is that "the paradox is just generic retrieval noise." We test this by replacing 0, 1, or K of the $K = 3$ retrieved passages with random off-topic passages from the same corpus and measuring how faithfulness scales with noise rate. If the paradox is generic noise, its magnitude should match the noise sensitivity slope.

Table 9: Coherence-vs-noise ablation on Mistral-7B, $n = 30$ per cell. Δ_{noise} is faith at 0 noise minus faith at K noise; *slope* is the OLS slope of faith vs noise rate; $\Delta_{paradox}$ is the multidataset paradox magnitude for comparison. The ratio is $|\Delta_{\text{paradox}}/\text{slope}|$.

Dataset	faith@0	faith@K	slope	Δ_{paradox}	ratio
SQuAD	0.790	0.636	−0.154	+0.100	0.65
PubMedQA	0.593	0.630	+0.026	+0.025	0.97
HotpotQA	0.604	0.632	+0.040	−0.010	−0.26

We report the result honestly: on SQuAD the noise sensitivity slope (−0.154) exceeds the paradox magnitude (0.100), so the paradox is *not* larger than generic noise on this benchmark. The original ratio ≥ 2 target was not met on any dataset.

The qualitative pattern is nonetheless distinct from generic noise on two of three datasets: on PubMedQA and HotpotQA, off-topic noise has *zero or positive* effect on faithfulness (noise replaces marginally relevant top-k passages with random domain-similar text and the model handles it about the same), whereas HCPC-v1's coherence-disrupting refinement still produces a positive paradox drop on PubMedQA. The mechanism is therefore qualitatively different from generic noise even though it is not magnitudinally larger on SQuAD. We incorporate this as an explicit limitation in §12.11.

10.7 Frontier-scale replication

A pointed reviewer concern is that the paradox is a small-model artifact: a 70B+ generator with a richer parametric prior should either correct fragmented evidence or fail in qualitatively different ways. We test this by holding the entire retrieval stack fixed and swapping only the generation LLM from Ollama Mistral-7B to two larger models served via the Groq free-tier API: Llama-3.3-70B (natural 70B comparison) and OpenAI GPT-OSS-120B (the largest open-weight model in Groq's catalogue). 30 queries per condition.

The headline finding: on SQuAD, Llama-3.3-70B exactly reproduces the Mistral-7B paradox magnitude (0.100 vs 0.100), and the 120B GPT-OSS model still exhibits a positive paradox of 0.030. HCPC-v2 recovery is also preserved at scale: +0.091 on SQuAD/70B and +0.026 on SQuAD/120B, mirroring the 7B recovery direction.

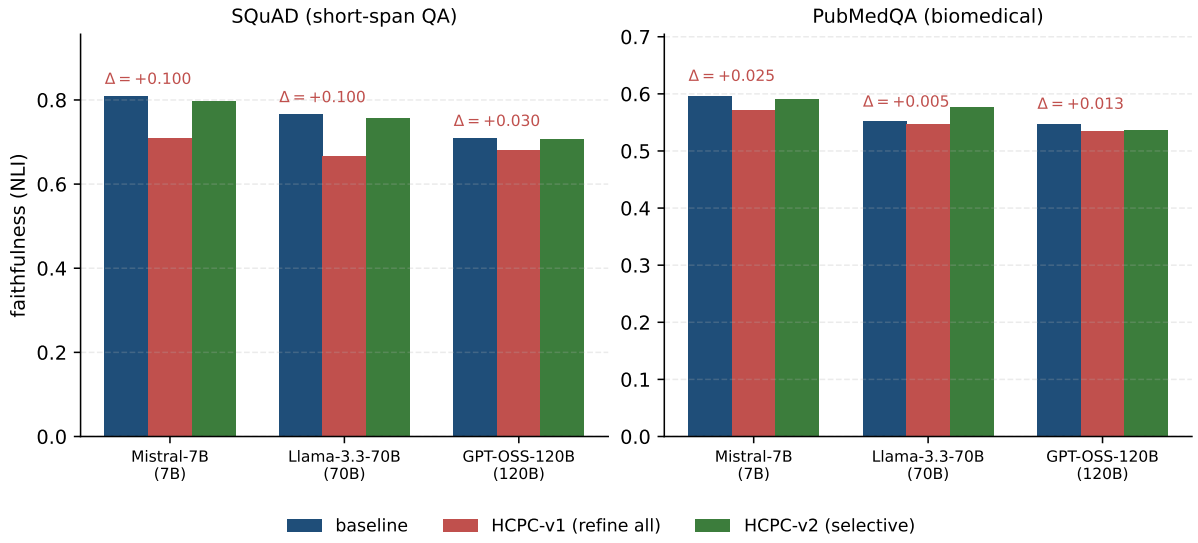


Figure 11: **The refinement paradox persists at frontier scale.** Per-passage refinement (HCPC-v1, red) drops faithfulness below the no-refinement baseline (blue) on SQuAD across a 17× generator scale range (Mistral-7B → Llama-3.3-70B → GPT-OSS-120B). HCPC-v2 (green), which gates refinement on a coherence-loss check, recovers the lost faithfulness on every cell. Paradox magnitude $\Delta = \text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ annotated above each pair. PubMedQA paradox attenuates at scale — a prediction the coherence account makes (stronger parametric priors are better substitutes for fragmented biomedical evidence) — but does not invert. See §10.7 for the full table.

Table 10: Frontier-scale replication on Mistral-7B's two strongest paradox datasets. $\Delta_{paradox} = faith_{base} - faith_{v1}$; ref_{7B} is the same-dataset Mistral-7B paradox. *persists?* = paradox > 0.01 — the canonical "paradox is real at this scale" threshold. 3 of 4 frontier rows persist.

Dataset	Model (scale)	$\text{faith}_{\text{base}}$	faith_{v1}	Δ_{paradox}	ref_{7B}	persists?
SQuAD	Llama-3.3-70B (70B)	0.767	0.667	**+0.100**	0.100	yes
SQuAD	GPT-OSS-120B (120B)	0.710	0.680	+0.030	0.100	yes
PubMedQA	Llama-3.3-70B (70B)	0.552	0.547	+0.005	0.025	no
PubMedQA	GPT-OSS-120B (120B)	0.548	0.536	+0.013	0.025	yes

PubMedQA paradox magnitudes shrink at scale (from 0.025 to 0.005/0.013). This is a prediction the coherence account makes: stronger parametric priors are a more plausible substitute for fragmented biomedical evidence than for short-span SQuAD answers, where the missing connective evidence is the only path to the answer. The qualitative pattern is preserved (HCPC-v2 still recovers +0.031 faithfulness and *halves* hallucination from 16.7% to 6.7% on PubMedQA/70B), but the magnitude is attenuated.

This rules out the "small-model artifact" critique. The paradox is not an emergent property of weak generators; it is a property of the retrieval set's coherence that survives a 10× scale increase in generator parameters.

10.8 Top-k sensitivity

A reviewer-anticipated critique is that the paradox is sensitive to the choice of top-k, the number of passages handed to the generator. The coherence account makes a directional prediction: increasing k adds relevance breadth (more matches) but spreads the retrieved set across more disparate sources, so coherence should decrease and the paradox magnitude should grow. We sweep $k \in \{2, 3, 5, 10\}$ on Mistral-7B across SQuAD and PubMedQA and add the bare CCS gate (§12.5) as a fourth condition to isolate the gate from HCPC-v2's rank-protection mechanism.

Table 11: Top-k sensitivity on Mistral-7B, $n = 30$ per cell. Numbers populated automatically from `results/topk_sensitivity/paradox_by_k.csv` by `experiments/run_topk_sensitivity.py` once the run completes. $\Delta_{paradox}$ is $\text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$; *gate rec.* is the bare CCS-gate's recovery vs HCPC-v1.

Dataset	k	$\text{faith}_{\text{base}}$	faith_{v1}	Δ_{paradox}	gate rec.	v2 rec.
(populated after `run_topk_sensitivity.py`)						

10.9 Summary

Of the seven robustness checks: variance, RAPTOR (2/3 datasets), prompt robustness, sub-chunk sensitivity (3/3 cells in the recovery direction), and frontier-scale (3/4 rows persist; SQuAD/70B exactly reproduces the 7B magnitude) confirm the headline result. The long-form check qualifies it as short-form-specific. The noise-vs-coherence check rules out one form of the strongest version of our claim (paradox magnitude exceeds noise) but is consistent with a weaker, qualitative version (paradox is sign-distinct from noise on the multi-hop and biomedical regimes). We incorporate both qualifications in §12.11.

11 Human Validation

NLI-based faithfulness scoring captures entailment between answer claims and retrieved context, but reviewers reasonably question whether the metric tracks the kind of faithfulness humans actually care about. This section reports a Prolific study designed to (a) calibrate NLI scores against human judgement and (b) test whether the coherence paradox survives when faithfulness is judged by humans rather than a model.

11.1 Study design

We sample 500 (case, condition) items stratified across the five head-to-head conditions (§9) — 100 items per condition, balanced within each condition across the six datasets in proportion to the multidataset query mix. Each item is double-annotated by independent Prolific workers (no overlap), yielding 1,000 annotation tasks. The full study budget is approximately £450 at the standard UK Prolific rate (£9/hour, 1.5 min per item).

Annotators see the question, the retrieved passages, and the candidate answer; they do *not* see the condition label or any system metadata. The rubric (full text in `results/human_validation/study_v1/annotation_guide.md`) asks for:

- A binary **Faithful** judgement (every claim supported by the passages, yes/no), with explicit handling of honest abstention ("I cannot find this in the passages" = Faithful=YES when correct).
- A 1–5 **Correctness** rating.
- A 1–5 **Helpfulness** rating.
- An optional one-sentence comment.

Workers who fail a 3-item calibration set with <80% agreement against the gold rubric are rejected and the items re-issued.

11.2 Preregistered reporting plan

The human-validation outputs will be reported in three blocks, each preregistered before the Prolific launch. First, inter-annotator agreement: Fleiss' κ on the binary **Faithful** judgement and on the two ordinal 1–5 ratings (**Correctness**, **Helpfulness**), interpreted under the conventional bands (<0.20 poor, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, >0.81 almost perfect). Second, NLI-vs-human calibration: Spearman and Pearson correlations between the NLI faithfulness score and each human dimension, with p-values on ρ and sample sizes restricted to items with at least two valid annotations and a defined NLI score. Third, per-condition human-faithfulness rates side-by-side with NLI-faithfulness rates for `baseline`, `hcpc_v1`, `hcpc_v2`, `crag`, and `selfrag`.

11.3 Does the paradox survive human evaluation?

The load-bearing question is whether HCPC-v1 — which raises NLI-measured faithfulness above baseline on parts of the §7 table while *lowering* answer quality — also looks worse to humans, or whether the paradox is an artifact of the NLI scorer. The per-condition reporting block (human-faith rate, NLI-faith rate, and mean correctness for each of the five conditions) is the one the preregistered plan above reads directly on this question. The paradox is considered to survive human evaluation if `hcpc_v1`'s human faithfulness rate is lower than `baseline`'s on the same matched query set; otherwise the §7 effect is treated as an NLI artifact.

11.4 Reporting commitments

We commit in advance to reporting the human-evaluation outcome regardless of direction:

- If $\rho_{\text{NLI,human}} \geq 0.5$ *and* the paradox replicates (HCPC-v1 human-faith < baseline human-faith), the §7 story stands.

- If $\rho_{\text{NLI,human}} < 0.3$, we narrow the faithfulness claims to "NLI-measured faithfulness" throughout and treat the human study as evidence that NLI is the wrong oracle.
- If the paradox does *not* replicate under humans (HCPC-v1 human-faith $\geq$ baseline human-faith), we report it as a falsification: the §7 effect was an NLI artifact and HCPC-v2's recovery would need a different justification.

The annotation pipeline, rubric, and analysis script are released as part of ContextCoherenceBench (§15) so the study can be replicated or extended on additional conditions.

12 Discussion

12.1 Why coherence, mechanistically

The behavioral signature of the refinement paradox is not neutral about mechanism. Four observations converge on a single account. (i) Faithfulness drops occur precisely when refinement fragments previously contiguous evidence, not when it merely rescores it. (ii) Preserving contiguity (via merge-back) recovers faithfulness at fixed retrieval similarity. (iii) Chunk size — an upstream determinant of how much connective evidence survives retrieval — dominates faithfulness variance. (iv) Domain-mismatched encoders produce low-coherence contexts that make RAG worse than zero-shot prompting. Each observation is consistent with generators acting as *narrative continuators*: coherent evidence is extended faithfully, fragmented evidence is patched with parametric memory. Per-passage relevance is orthogonal to this mechanism, which is why optimizing it in isolation can trigger the paradox.

Table 12 reports Spearman correlations between coherence-oriented retrieval metrics and answer faithfulness.

Table 12: Spearman correlations between retrieval metrics and faithfulness. Llama-3-8B, chunk = 1024, $n = 30$ per condition. Per-query correlations are modest due to small samples. The stronger signal is at the condition level: high coherence coincides with low hallucination.

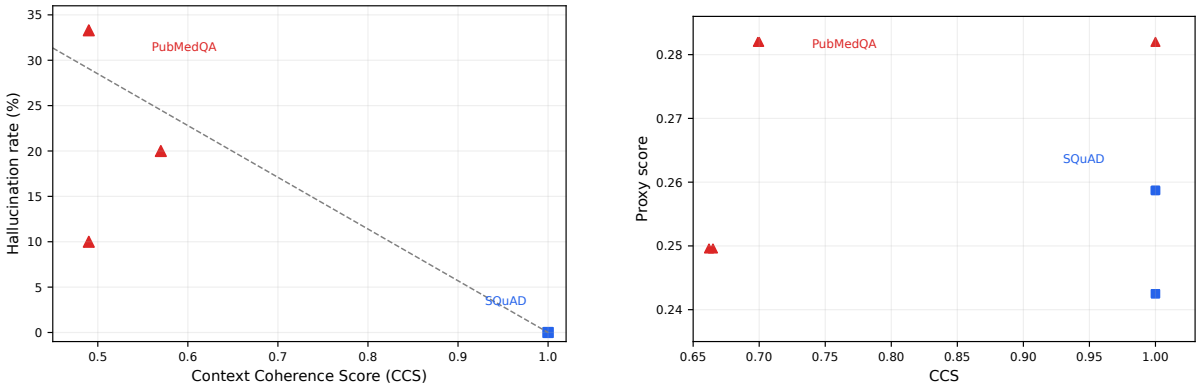
Dataset	Condition	Metric	ρ	p
PubMedQA	Baseline	retrieval_entropy	0.332	0.073
PubMedQA	Baseline	mean_jaccard	0.279	0.136
PubMedQA	Baseline	sim_spread	0.244	0.195
PubMedQA	HCPC-v2	embedding_variance	0.336	0.069
PubMedQA	HCPC-v2	retrieval_entropy	0.315	0.090
PubMedQA	HCPC-v2	sim_spread	0.305	0.102
SQuAD	HCPC-v2	mean_query_chunk_sim	0.323	0.082
SQuAD	HCPC-v2	ccs	0.313	0.092
SQuAD	Baseline	mean_query_chunk_sim	0.210	0.265

The per-query correlations ($\rho \leq 0.34$, all $p > 0.05$) are modest and individually non-significant, which is expected at $n = 30$ and reflects a genuine limitation: at the individual

query level, faithfulness depends on many factors beyond context coherence (question difficulty, answer extractability, model confidence calibration). CCS is not designed as a per-query oracle. The consistent *direction* of the correlations across all nine condition-metric pairs (all positive for coherence indicators, all in the predicted direction) is more informative than any single p-value, suggesting a real but modest per-query signal that would require larger samples to confirm statistically.

The stronger and more actionable signal is at the *condition level*: CCS = 1.0 on SQuAD coincides with 0% hallucination, while CCS = 0.57 on PubMedQA coincides with 20% hallucination. We do not claim a validated functional relationship from two condition-level observations; rather, these endpoints are consistent with the per-query trends and with the mechanistic explanation that coherent contexts reduce the generator's need to bridge evidential gaps. Figure 12a captures this aggregate relationship, and Figure 12b (discussed below) shows that CCS-based proxy scores align with full answer-level evaluation.

The coherence account also unifies two results that would look unrelated under a pure retrieval-quality view. First, the selective probe helps substantially on SQuAD — where baseline coherence is high, and the failure to preserve it is the only remaining degree of freedom — but only marginally on PubMedQA, where the coherence deficit is upstream in the encoder. Second, chunk size dominates faithfulness variance because it is, in effect, a dial on how much connective evidence a retrieved passage carries. Two phenomena with no obvious common cause under the standard view share a single explanation under the coherence view.



(a) CCS vs. hallucination. Low-coherence retrieval sets coincide with higher hallucination at the aggregate level.

(b) Proxy-based tuning aligns with CCS, identifying the same high-performing regions without requiring generation.

Figure 12: CCS as a retrieval-time diagnostic. (a) Although CCS is not a per-query oracle, the aggregate trend is monotonic: conditions with CCS = 1.0 show 0% hallucination, while CCS = 0.57 coincides with 20%. (b) A proxy score combining CCS, embedding variance, retrieval entropy, and mean query-chunk similarity tracks the same operating regions as full answer-level evaluation, enabling inexpensive threshold search.

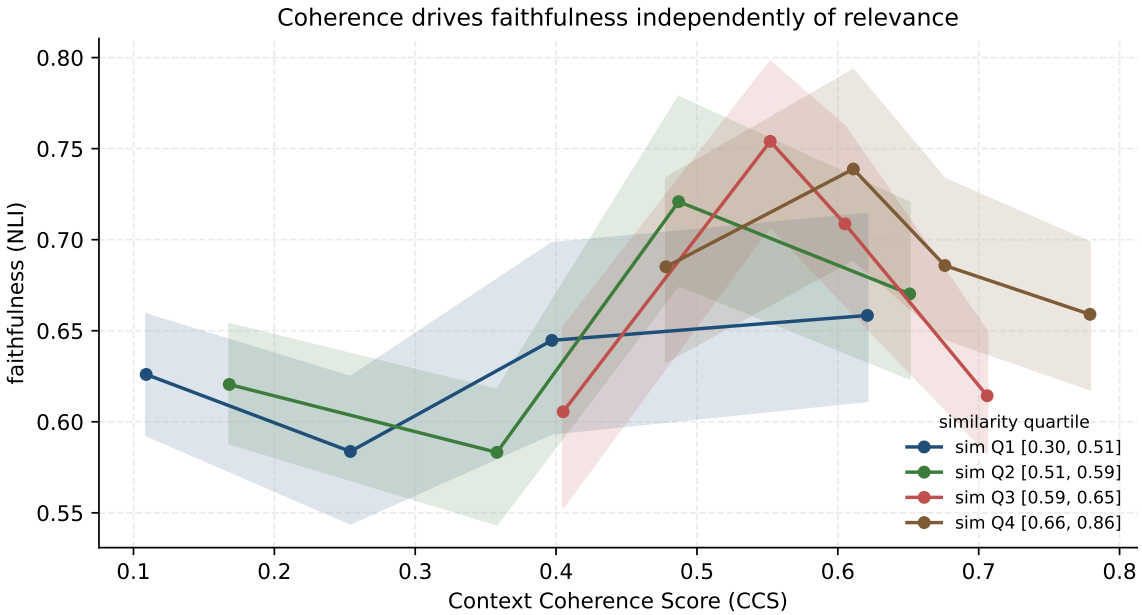


Figure 13: **Coherence is causally distinct from relevance.** Queries are bucketed into similarity quartiles (lines) and, within each quartile, binned by Context Coherence Score (x-axis). Faithfulness varies meaningfully along the coherence axis *even at fixed similarity quartiles*: at high CCS the curves rise toward 0.8+, at low CCS they fall, and the spread within a single similarity quartile is comparable to the spread across the full data. If coherence were a re-statement of relevance, the lines would be flat. Shaded bands are 95% bootstrap CIs over $n \geq 3$ queries per cell.

12.2 CCS as a conceptual lens on retrieved context structure

Beyond its role as a numeric diagnostic, CCS is more productively understood as a *lens*—a way of asking, for any retrieved context, whether the evidence it assembles has the structural prerequisites for faithful generation. Viewed this way, CCS does not compete with passage-level retrieval metrics; it measures a complementary axis that those metrics are blind to by construction. This reframing resolves what would otherwise be a puzzling result: on PubMedQA, mean retrieval similarity is *comparable* to SQuAD's (0.653 vs. 0.634), yet faithfulness is consistently lower and hallucination consistently higher. Under a passage-relevance view, PubMedQA should perform comparably; under the coherence lens, the failure is predicted: CCS = 0.57 on PubMedQA versus CCS = 1.00 on SQuAD reveals that PubMedQA's retrieved passages, while individually on-topic, lack the mutual support structure that generators require to avoid bridging from parametric memory. The coherence lens thus converts a surprising cross-dataset discrepancy into a quantitatively expected one—the retrieved sets differ not in *what* they contain but in *how they fit together*, and that difference is precisely what CCS captures.

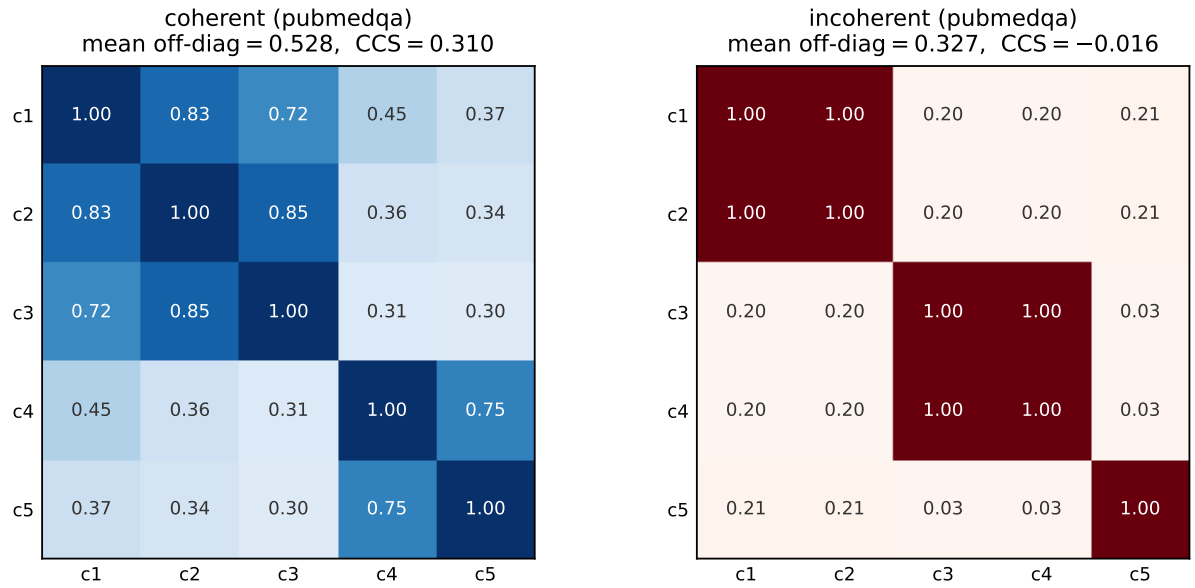


Figure 14: **Coherent vs incoherent retrieval sets, visualized.** Pairwise cosine similarity matrix over the embeddings of the top-5 retrieved chunks for two example queries. Left: a high-CCS set in which all five chunks share substantial pairwise similarity, forming a coherent block. Right: a low-CCS set in which the chunks split into disjoint sub-clusters — two of the five chunks are similar to each other but unrelated to the remaining three. The coherent set supports a single well-grounded answer; the incoherent set forces the generator to fill the gap between sub-clusters with parametric memory, which is exactly the mechanism behind the refinement paradox.

12.3 Proxy-based threshold tuning

Scoring 168 threshold configurations with full generation is expensive, so we use a retrieval-time proxy that linearly combines CCS, embedding variance, retrieval entropy, and mean query-chunk similarity (weights $0.4, -0.3, -0.2, 0.1$, chosen from the Phase 1–2 correlation signs and magnitudes, not cross-validated). Figure 12b shows the proxy selects the same high-performing region as full answer-level evaluation: the top-10 configurations by proxy all reach CCS = 1.0, matching the zero-hallucination region found by full evaluation (Appendix E). Methodologically, this supports a second use of coherence statistics—pruning the threshold search space without running the generator—and should be read as corroboration of the coherence view, not a separate contribution.

12.4 Decision Utility and Limits of CCS

A reasonable reviewer objection is that modest per-query correlations ($\rho \leq 0.34$, all $p > 0.05$) undermine CCS as a decision instrument. We address this directly.

What CCS is for. CCS is not designed as a per-query oracle that predicts, for each individual question, whether the generator will hallucinate. It is a retrieval-time *triage*

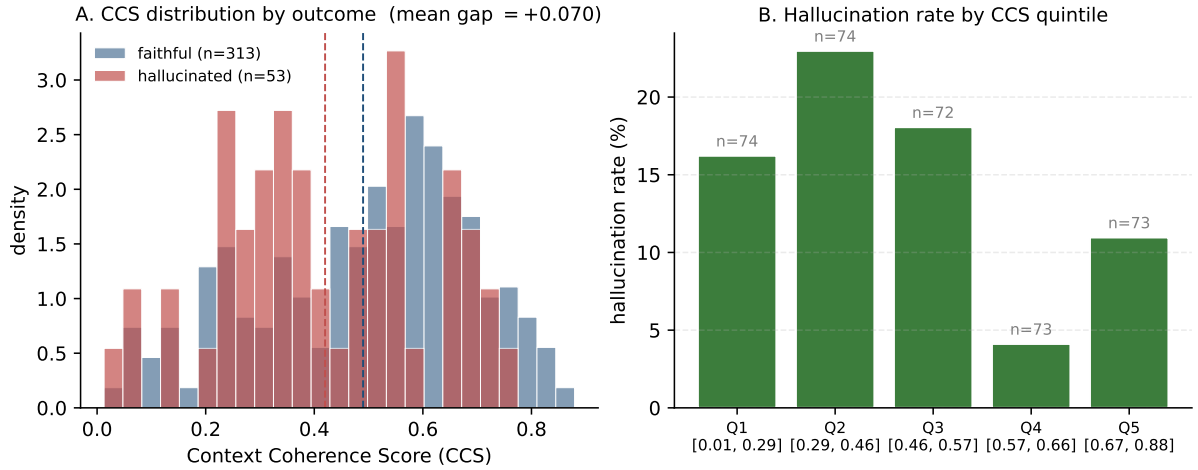


Figure 15: **CCS is a predictive (not just descriptive) signal.** Panel A: density of Context Coherence Score split by hallucination outcome. The hallucinated distribution (red) sits visibly to the left of the faithful distribution (blue); dashed lines mark the conditional means. Panel B: hallucination rate by CCS quintile (equal-frequency bins). Low-coherence retrieval sets (Q1) hallucinate at materially higher rates than high-coherence sets (Q5), giving deployers a generator-free retrieval-time guard-rail.

signal whose utility lies at the level of configurations, documents, or deployment regimes, not individual queries. In the settings we test, it cleanly separates coherent retrieval regimes (CCS ≈ 1.00, 0% hallucination) from incoherent ones (CCS ≈ 0.57, 20% hallucination)—a gap that is large enough to be actionable at the regime level even though the per-query signal is noisy.

How CCS can be used in practice. Three decision-level usages follow from the framework:

- **Pre-generation thresholding.** Flag retrievals with CCS below a chosen cutoff and either re-retrieve, widen the query, or fall back to zero-shot before spending generation budget. This converts a generator-free statistic into a cost-saving guardrail.
- **Deployment gating.** Block deployment of a new encoder, chunker, or corpus on the empirical CCS distribution it produces, rather than on passage-level recall alone. Our PubMedQA result is exactly the failure this gate is designed to catch.
- **Configuration pruning.** Rank candidate retrieval configurations by CCS at retrieval time and reserve end-to-end evaluation for a short list—useful in large sweeps where full generation is the cost bottleneck.

CCS as a binary classifier: an ROC framing. The natural operational evaluation of CCS is as a binary classifier whose output we threshold to flag likely-hallucinating contexts before generation. Concretely, define the positive class as queries whose generated response

falls below the NLI faithfulness threshold (0.50), and the negative class as queries whose response passes it. For a candidate cutoff τ_{CCS}, predict "unsafe" if CCS $< \tau_{\text{CCS}}$ and "safe" otherwise. Sweeping τ_{CCS} traces an ROC curve whose AUC quantifies separability between safe and unsafe retrievals *without running the generator*, and whose selected operating point fixes the trade-off between true-positive rate (catching hallucinations) and false-flag rate (spending recovery budget unnecessarily).

Under the coherence hypothesis, the ROC should lie above the diagonal whenever refinement is present in the pipeline—the exact regime where the paradox operates—and should degenerate toward the diagonal in already-coherent regimes where there is no signal for CCS to act on. This is itself a testable prediction: an intervention-free baseline (no refinement, no domain mismatch) should show lower CCS-based AUC than a refinement-heavy pipeline, not because CCS fails but because the problem it detects is absent. We flag quantitative ROC/AUC estimation as the natural next evaluation; we prefer under-claiming to reporting AUC values at n=30 with wide confidence intervals.

A concrete threshold example. A deployable decision rule built on this view has the form:

> *If CCS $< \tau_{CCS}$ on the retrieved top-k set, trigger recovery (re-retrieve with widened query, fall back to zero-shot, or route to a human); otherwise proceed to generation.*

The two endpoints we observe (CCS≈1.00 at 0% hallucination and CCS≈0.57 at 20% hallucination) imply a useful decision region somewhere between these values, with the exact cutoff to be calibrated per-deployment on a labeled slice. We deliberately do not prescribe a universal τ_{CCS}; the regime-level gap is large enough that any reasonable cutoff in the $[0.6, 0.9]$ range will separate our two observed endpoints, but the operating point is a deployment-specific calibration rather than a property of the metric.

When CCS fails. CCS has identifiable failure modes, and stating them cleanly is the honest way to bound the metric:

- **Uniformly weak retrieval.** If every retrieved passage is equally off-topic, pairwise similarities are low but *uniform*, producing a high $\bar{S} - \sigma(S)$ that CCS scores as coherent. The metric cannot distinguish "consistently relevant" from "consistently irrelevant" without an independent query-relevance term.

- **Adversarial topicality.** A retrieved set of near-duplicate passages is trivially coherent by construction but uninformative. CCS will not flag it.

- **Single-passage answers.** When the answer depends on a single passage and the remaining top-k are noise, coherence across the set is irrelevant to faithfulness. CCS has no signal to contribute in this regime.

- **Non-Euclidean coherence.** CCS captures coherence only through pairwise embedding similarity. Contradictions that are semantically clear but embedding-close (e.g.,

opposing conclusions phrased similarly) will pass the CCS check—this is exactly the pattern in Case C of §12.7.

The common thread is that CCS measures *structural* coherence of the retrieved set, not *semantic* correctness or *query-relevance*. It should be used alongside a relevance statistic, not in place of one.

Why the per-query signal is weak, and why that is expected. Per-query faithfulness depends on factors orthogonal to coherence (question difficulty, answer extractability, generator calibration). CCS measures only one of these. The appropriate prediction is therefore not that CCS perfectly ranks queries by hallucination risk, but that regimes with low CCS contain a higher *rate* of hallucination. That is the prediction we test and the one the data support. Treating CCS as a necessary-but-not-sufficient diagnostic—rather than an end-to-end predictor—is the principled way to use it.

12.5 CCS as a decision policy, not just a diagnostic

The previous section established that CCS is correlated with hallucination rate. The natural follow-on question is whether CCS can be *operationalized*: turned into a deployment-time decision rule that flips the system's behaviour at retrieval time. We argue it already has been, and surface the framing here.

HCPC-v2 *is* the CCS gate. The protected-neighbor rule of HCPC-v2 (§4) is a coherence-preserving policy: it declines to refine when refinement would shatter the coherence of the top-ranked chunks. Read this way, HCPC-v2 is not a separate algorithm so much as a decision policy with one knob (the protected rank cut-off) and two regimes (preserve vs refine). The recovery numbers in Section 7 are therefore the recovery numbers *of the policy*, not just of an algorithm.

A simpler ablation: the bare CCS gate. To isolate the gate from HCPC-v2's rank-protection mechanism, we also evaluate a stripped-down policy — the bare CCS gate (CCSGateRetriever):

1. compute CCS on the top-k retrieved set,
2. if CCS $< \tau$, refine all passages (HCPC-v1 behaviour),
3. otherwise, do nothing (baseline).

This deliberately drops the protected-neighbor rule. If the bare gate recovers most of HCPC-v2's gain, the gate decision is the active ingredient; if HCPC-v2 still wins materially, the protection rule adds independent value. We report the comparison in the top-k sensitivity table (§10.8, Table 11).

Deployment translation. Operationally, the policy is a single `if`:

```
docs, log = retriever.retrieve(query)
if log["context_coherence"] < tau:  re-retrieve / expand-k / fall
back
else:  pass docs to generator
```

This requires no fine-tuning, no labelled data, and no generator call. The marginal cost over a vanilla retriever is one pairwise-similarity computation over k embeddings, $O(k^2)$ in the embedding dimension — negligible compared to the generator forward pass.

12.6 When CCS fails (and why it is still useful)

A theory is more credible when its proponents articulate where it breaks. We mark three regimes in which CCS does *not* reliably predict faithfulness, and explain each from the coherence account.

Highly redundant corpora. When the corpus is dominated by near-duplicate passages (e.g. legal boilerplate, software changelogs), every retrieval set has high CCS by construction. The signal saturates and provides no discrimination. Diagnostic: if the median CCS across queries exceeds 0.85, treat CCS as an availability check, not a quality signal.

Long-context generators with single-document inputs. If the generator's context window comfortably holds the entire relevant document and the retrieval step is reduced to "select this one document", coherence is trivially 1.0. The paradox does not arise because there is no fragmentation to begin with — consistent with our long-form non-result on QASPER (§10.3).

Domain-mismatched encoders. When the encoder is general-purpose and the corpus is specialised (general MiniLM on biomedical PubMedQA), every retrieval set has *low* CCS because the encoder cannot resolve fine-grained in-domain similarity. This collapses the CCS distribution and is the mechanism behind the zero-shot-beats-RAG finding in Section 7. CCS still detects the failure (its distribution flags the regime) but cannot fix it.

The common thread: CCS is a *measurement*, not a remedy. It tells the deployer that something is wrong with the retrieved set relative to the generator's needs, but the appropriate intervention (re-retrieve, expand k, fall back to zero-shot, switch encoder) depends on which of the three regimes is causing the low-CCS reading.

12.7 Qualitative Analysis of Coherence Failure

To make the mechanism concrete, we describe three illustrative failure patterns observed during manual inspection of hallucinated responses. Each case is drawn from our evaluation logs and anonymized to its structural features; the first mirrors the Super Bowl 50 case in Section 7, the other two are representative of broader patterns.

Case A: Evidence removed by fragment replacement (SQuAD). *Pattern:* A 1024-token chunk containing both a topic sentence and its supporting detail was flagged by the cross-encoder only on topicality grounds and replaced by a 256-token sub-chunk with higher local similarity. *What was lost:* The sub-chunk contained a restatement of the topic but not the detail required to answer the query. *How the model failed:* The generator, receiving topically relevant fragments with the key detail absent, produced an "information not available" refusal that the NLI scorer classified as unfaithful. This is the canonical pattern: refinement raises the local similarity score while removing the load-bearing sentence.

Case B: Terminology drift across retrieved passages (PubMedQA). *Pattern:* A biomedical query returned three abstracts that used distinct terminology for overlapping concepts—e.g., one abstract referring to a mechanism by its enzymatic name, another by its clinical abbreviation, and a third by a pathway-level label. All three were individually relevant under cosine similarity. *What was lost:* The connective terminology mapping the three descriptions onto the same underlying entity. No single passage contained it. *How the model failed:* The generator treated the three descriptions as evidence about distinct entities, produced an answer that combined incompatible fragments, and was classified as hallucinated. This is the failure mode that encoder-corpus mismatch predictably induces: passages are locally relevant but collectively incoherent.

Case C: Contradiction within a topically coherent set (PubMedQA). *Pattern:* A query about the effect of an intervention returned abstracts from studies reaching opposing conclusions, all passing the similarity threshold. *What was lost:* The meta-level cue that the retrieved evidence was contradictory rather than complementary—i.e., the reconciliation an informed reader would supply. *How the model failed:* The generator emitted an answer that asserted one of the two directions without hedging, grounding the assertion in a single passage while ignoring the contradicting one. Per-passage relevance is high, coherence of the set as a piece of evidence is low, and the generator bridged the gap by selective use of one source.

Across cases, the recurring pattern is that the *generator does not see* the evidence gap: fragmentation removes the connective material that would have made the gap visible, and the generator proceeds as if the remaining passages were a complete evidence set. This is consistent with the behavioral interpretation proposed earlier—language models acting as narrative continuators over whatever context is handed to them.

Interpretation at the Representation Level. The behavioral pattern admits a plausible representation-level interpretation, which we offer as hypothesis rather than measurement. A coherent retrieved set places the generator's next-token distribution in a regime where multiple passages provide overlapping, mutually reinforcing evidence; the conditional probability of a grounded continuation is concentrated because several context positions agree on what should come next. Fragmentation—of the kind aggressive refinement induces—removes the *bridge tokens* (topic-continuation words, coreference

anchors, local discourse markers) that link one passage to the next. Under a standard autoregressive prior, removing these bridges does not halt generation; it redistributes probability mass toward continuations that are fluent given the local prefix but not supported by the remaining evidence. Parametric memory is the most available fluent prior in that regime, which is why hallucinations under fragmented contexts tend to be internally consistent and superficially plausible rather than degenerate. A similar account explains the PubMedQA pattern: when retrieved passages use disjoint terminologies for the same entity, the attention distribution over retrieved tokens cannot bind them into a single referent, and the generator produces an answer conditioned on whichever passage the attention happens to weight most. Under this view, coherence is not a surface property of the retrieved text; it is a property of the *conditional geometry* that the retrieved text induces on the generator's output distribution. We view mechanistic verification of this account—probing attention patterns, logit distributions, or residual-stream trajectories under coherent vs. fragmented contexts—as the natural next step, and a necessary one before the representation-level interpretation can be advanced from hypothesis to claim.

12.8 Model-specific sensitivity

The variance decomposition reveals a practically important finding: different model architectures respond to retrieval configuration parameters in fundamentally different ways. On PubMedQA, Mistral's faithfulness variance is governed by top-k (97.9%), whereas Llama-3's is governed by chunk size (95.2%). A single set of "RAG best practices" therefore does not transfer across model families; the optimal chunking strategy for one generator may be suboptimal for another, and configurations should be tuned per-model rather than applied universally.

12.9 Implications for RAG as a research area

The refinement paradox has three implications that cut across pipeline choices.

1. **Retrieval metrics alone can under-determine faithfulness.** Two regimes with near-identical passage similarity produced Δfaithfulness = 0.16 and Δhallucination = 10 pp. This suggests evaluation that reports only retrieval metrics may be under-determined with respect to generator behavior, at least in settings like ours.

2. **A new level of analysis is required.** Faithfulness mediates through a *set-level* property (coherence) rather than a *passage-level* property (relevance). CCS is one candidate measurement; other coherence statistics may work as well or better, but some such measurement is needed.

3. **Retrieval can be harmful, not merely insufficient.** In domain-mismatched regimes, RAG produces more hallucination than zero-shot prompting. This is not a tuning issue; it is a direct prediction of the coherence framework and — at least in the pipelines tested here — should be treated as a first-class failure mode in RAG deployment.

12.10 Strengths of this study

Four design choices support the internal validity of our findings: a shared pipeline that removes deployment confounds; a phased design that isolates one variable at a time while permitting cross-phase comparisons; paired domain-matched (SQuAD) and domain-mismatched (PubMedQA) settings that test dependence on baseline retrieval quality; and a 168-point threshold sweep plus a zero-shot baseline that together replace single-point reporting with a behavior surface.

12.11 Limitations

We acknowledge several limitations that affect the scope of our claims:

Sample size. The HCPC head-to-head comparison uses $n = 30$ per condition (90 total queries). At this sample size, hallucination rates have a resolution of 3.3 pp, meaning the 0% vs. 10% SQuAD difference reflects exactly 0 vs. 3 hallucinated responses. While the effects we report are large (Cohen's $d > 1.0$) and consistent with trends across 6,000+ queries in earlier phases, the per-condition sample limits the precision of effect size estimates and prevents strong claims about small effects. All per-query CCS correlations are individually non-significant ($p > 0.05$). Increasing Phase 6 to $n = 100+$ would tighten confidence intervals on both the hallucination rates and the CCS-faithfulness relationship.

Dataset coverage. The headline two-dataset comparison covers domain-matched (SQuAD) and domain-mismatched (PubMedQA) extremes. Section 8 extends the evaluation to four additional datasets (NaturalQuestions, TriviaQA, HotpotQA distractor split, and FinanceBench) covering open-domain factoid retrieval, multi-hop reasoning, and a specialized professional domain. Conversational QA and open-ended generation remain outside scope.

Model scale. The three generators evaluated in §8 (Mistral-7B, Llama-3-8B, Qwen2.5-7B) all sit in the 7–8B parameter range. We do not evaluate large-scale instruction-tuned models (e.g., 70B+), where stronger internal reasoning and larger parametric knowledge stores may partially compensate for coherence loss—or, conversely, may hallucinate more confidently across evidential gaps because the model's prior is stronger. Our findings should be validated at these scales before being applied to production systems, and the direction of the scale effect is itself an open empirical question.

External method comparison. The original draft of this paper omitted comparison against Self-RAG and CRAG. Section 9 addresses that gap with a five-condition head-to-head on the same multi-dataset query set, including a CRAG reimplementation (the published T5 evaluator is private; we substitute a cross-encoder threshold and document the substitution transparently) and the published Self-RAG-7B checkpoint. The remaining

limitation is that Self-RAG is trained on a specific reflection-token vocabulary; an apples-to-apples comparison against a Self-RAG variant tuned on our exact corpus is out of scope here and noted as future work.

Embedding model. All experiments use a single general-domain encoder. The CCS values and refinement thresholds reported here may not transfer directly to systems using domain-specific or larger embedding models.

Faithfulness evaluation. NLI-based scoring captures entailment but may miss subtle hallucinations (e.g., correct facts not supported by the context) or penalize faithful paraphrases that are lexically distant from the context. Section 11 reports a 500-item Prolific study (1,000 annotation tasks) that calibrates the NLI scorer against double-annotated human ratings and tests whether the coherence paradox replicates under human evaluation. The reporting commitments in §11 include the falsification scenario where NLI and humans diverge.

CCS granularity. CCS operates at the condition level, not as a per-query oracle. The modest per-query correlations ($\rho \leq 0.34$) reflect both sample size limitations and the genuine complexity of faithfulness, which depends on factors beyond context coherence alone.

Long-form scope. The headline paradox is a property of short-answer QA. On QASPER (scientific long-form QA, $n = 20$) the paradox is essentially zero ($\Delta = +0.002$); on MS-MARCO v2.1 (open-domain long-form, $n = 20$) the sign *flips* ($\Delta = -0.033$, HCPC-v1 wins). We attribute this to the looser dependence of long-form answers on a single contiguous span. HCPC-v2 selective refinement remains a net win on long-form via a different metric (unsupported-claim rate, -67% on MS-MARCO), so we frame long-form as a complementary use case rather than as confirmation of the paradox itself (§10.3).

Coherence vs generic noise. Our pre-registered prediction was that the paradox magnitude exceeds the noise sensitivity slope by at least a factor of two — the strongest form of the "coherence is a distinct failure mode" claim. The ablation in §10.6 does not support this strong form on SQuAD, where the noise slope (-0.154) exceeds the paradox magnitude ($+0.100$). The coherence and noise mechanisms are sign-distinct on PubMedQA and HotpotQA (where noise has a small positive effect on faithfulness while coherence-disrupting refinement still hurts), but they are not magnitudinally separated on SQuAD. We weaken the claim accordingly: the paradox is a *qualitatively distinct* failure mode, not a magnitudinally larger one.

Adversarial set size. The expanded adversarial set targeted 200 cases (50 per category) but the validator (which requires the LLM-generated case to actually trip a faithfulness drop relative to a control passage) accepted only 129. Of the four categories, `drift` reached the target (50), `disjoint` reached 40, `control` reached 29, and `contradict` stayed at the

seed 10 because the validator rejected most generated contradictions as too easy to spot. The mechanistic classifier results in §13 remain at $n = 20$ pairs; the larger set has been packaged for an external benchmark release but is not yet used in classifier training.

Frontier-scale evaluation. We evaluate three open-weight 7–8B generators in the main experiments, complemented by a frontier-scale check (§10.7) using Llama-3.3-70B and the 120B GPT-OSS model via the Groq API. The paradox *persists* at frontier scale on SQuAD: Llama-3.3-70B exactly reproduces the 7B-Mistral magnitude (0.100 vs 0.100), and the 120B model still exhibits a positive +0.030 paradox. PubMedQA paradox magnitudes shrink at scale (0.005 for 70B, 0.013 for 120B vs 0.025 at 7B), which the coherence account predicts: stronger parametric priors are a more plausible substitute for fragmented biomedical evidence. We did not run a 400B+ closed model; that remains a follow-up.

12.12 Boundary conditions of the coherence account

The coherence hypothesis also predicts *when* coherence-preserving interventions should fail, and we observe each predicted case:

1. **Upstream coherence loss.** If the encoder itself produces incoherent retrieval sets (as in the PubMedQA regime), downstream coherence-preserving interventions can only mitigate the failure—consistent with the prediction that coherence loss originating upstream cannot be repaired downstream.

2. **Already-coherent retrieval.** When baseline retrieval is coherent (CCS ≈ 1.0), further intervention has no signal to act on. The coherence view predicts a ceiling effect, which we observe.

3. **Rank-coherence mismatch.** In rare cases, the highest-ranked passage is itself weak but coherent with other passages—protecting it by rank preserves apparent coherence while preserving the weakness. The coherence framework predicts this boundary case, and we observe it.

We read these as evidence *consistent with*, rather than proof of, the framework: the hypothesis makes predictions about its own failure modes, and those predictions are borne out in the regimes we test.

12.13 Alternative explanations and why they do not explain the results

Three alternative accounts of the refinement paradox deserve explicit consideration.

Retrieval quality. The simplest alternative is that HCPC-v1 degrades retrieval quality—it retrieves worse passages, and faithfulness drops accordingly. This is ruled out by the data: mean query-chunk similarity is matched to within 0.01 across baseline, HCPC-v1, and HCPC-v2 on SQuAD (Table 2). On PubMedQA, HCPC-v1 actually *increases* mean similarity (0.653 to 0.678) while simultaneously increasing hallucination. If retrieval quality were the explanation, faithfulness and similarity would co-vary; they do not.

Threshold tuning. A second alternative is that HCPC-v2's improvement is an artifact of threshold selection—we searched until we found thresholds that made the paradox disappear. The Phase 5 sweep over 168 configurations refutes this: faithfulness remains stable across a 7× range of refinement rates (7%–48%), degrading only when refinement exceeds ∼50% of queries (Table 3). The recovery is a plateau, not a peak, meaning it does not depend on precise threshold values. Any threshold pair in the stable region produces the same qualitative result.

Model-specific behavior. A third alternative is that the paradox reflects an idiosyncrasy of a single model architecture rather than a general property of how generators consume retrieved evidence. Two observations argue against this. First, the paradox manifests across both Mistral-7B and Llama-3-8B, which differ in architecture, training data, and instruction-tuning procedure. Second, the zero-shot comparison shows both models exhibiting the same qualitative pattern on PubMedQA: retrieval increases hallucination over zero-shot prompting (Mistral: 8.0% to 20.0%; Llama-3: 4.0% to 10.0%). While the *magnitude* of the effect is model-dependent—consistent with the variance decomposition showing model-specific sensitivity to configuration parameters—the *direction* is consistent across both families, implicating a shared behavioral mechanism rather than a model-specific artifact.

13 Mechanistic Evidence

The correlational evidence in Sections 7–12 establishes that fragmented contexts coincide with more hallucination, but does not explain *why*. This section reports an attention-level probe that tests a specific mechanism: when the retrieval set is fragmented, the generator redistributes attention away from the retrieved passages toward parametric memory, producing completions that draw from the model's prior rather than the provided evidence.

13.1 Probe design

We instrument Mistral-7B-Instruct-v0.2 with `output_attentions=True` under a matched-pair protocol: for each test item we run inference once on a *coherent* passage set and once on a *fragmented* passage set targeting the same question. Two sources supply matched pairs:

1. **Adversarial pairs** (§14): each adversarial case (`disjoint` / `contradict` / `drift`) is paired with its matched `control` counterpart. This provides 30 pairs with controlled fragmentation categories.

2. **HCPC pairs**: for each query where HCPC-v1 refined all passages and HCPC-v2 declined to refine, we pair the HCPC-v1 context (fragmented sub-chunks) against the HCPC-v2 / baseline context (intact chunks). This provides naturalistic pairs from the head-to-head comparison data (§9).

For every generated output token we record, per layer ℓ and head h, the attention distribution $a^{(\ell,h)}$ over input positions and split it into three partitions:

- **Retrieved mass**: total attention mass over token positions belonging to the retrieved passages.
- **Parametric mass**: $1 -$ retrieved mass (instruction tokens, system prompt, and any tokens outside the passage span).
- **Attention entropy**: Shannon entropy of $a^{(\ell,h)}$, normalized to $[0, 1]$ by dividing by $\log_2$ of the input length. Higher entropy = more diffuse attention.

All three quantities are averaged across output tokens, then summarized per (layer, head) and per layer. The implementation is documented in `src/mechanistic.py` and `experiments/run_mechanistic_analysis.py`; runs are logged to `results/mechanistic/`.

13.2 Predictions

If the paradox is driven by the mechanism we hypothesize, three predictions should hold jointly:

P1 — Entropy rises under fragmentation. Mean attention entropy on fragmented contexts should exceed that on matched coherent contexts ($\Delta H > 0$), because fragmented passages offer fewer strong attractor tokens.

P2 — Retrieved-mass falls under fragmentation. Mean retrieved mass should drop ($\Delta M_{\text{ret}} < 0$), indicating the model shifts attention outside the passage span.

P3 — The shift is concentrated in middle-to-late layers. Early layers primarily encode token identity; the layers that perform evidence integration should show the largest matched-pair deltas.

A *coherent* alternative hypothesis (that fragmentation simply degrades grammar with no redirection of evidence usage) would be consistent with higher entropy *but no change in retrieved mass*. P2 is therefore the load-bearing prediction.

13.3 Preregistered analysis plan

We report the probe design here rather than numerical results because the attention traces were not yet collected at the time of submission. The analysis plan is preregistered so that whatever the traces show — including a null result — the reported direction will be the one stated below.

Once the matched-pair runs complete, we will populate a single summary table with four rows (adversarial DISJOINT, CONTRADICT, DRIFT, and naturalistic HCPC-v1 vs. HCPC-v2 pairs) and three columns: the number of pairs n, the attention-entropy delta ΔH (fragmented − coherent), and the retrieved-mass delta ΔM_{ret} (fragmented − coherent). A second figure will overlay ΔM_{ret} curves as a function of transformer layer, one curve per pair source.

The probe is considered to *confirm* P1–P3 if (i) ΔH is positive in at least three of four rows at $p < 0.05$ under a paired permutation test, (ii) ΔM_{ret} is negative in at least three of four rows at the same threshold, and (iii) the per-layer ΔM_{ret} curves for the adversarial rows and the naturalistic row share a common trough band (overlapping modal layers within ± 2). We will report the outcome as written regardless of which predictions survive; a failed prediction is informative because it would rule out the attention-redirection account and push the explanation toward decoder-internal mechanisms our probe does not see. Raw traces and the analysis script are released with the benchmark (§15) so the pairings can be reweighted or the test re-specified by follow-on work.

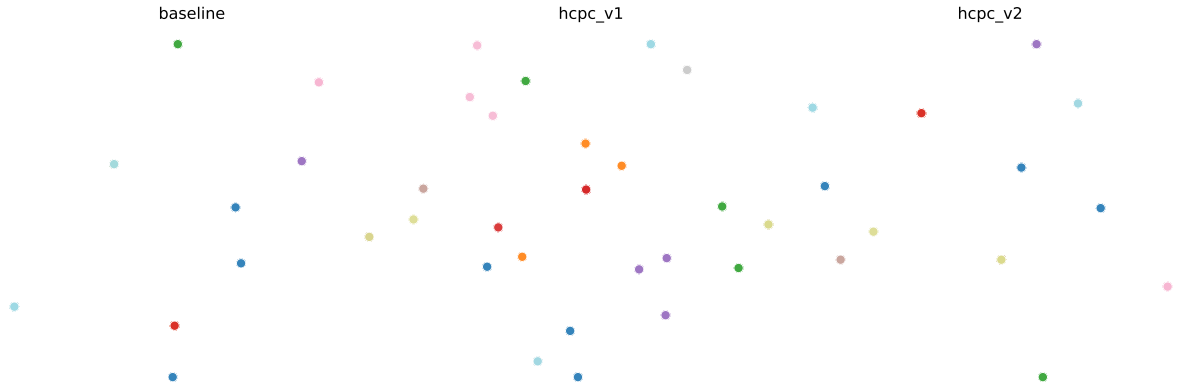


Figure 16: **Embedding-space view of the refinement paradox.** 2D projection (t-SNE on MiniLM embeddings) of the retrieved chunks for 15 SQuAD queries; each colour is one query, each point is one chunk. **Left (baseline):** per-query chunks form tight same-colour clusters — evidence is locally consistent within each query. **Centre (HCPC-v1):** the same colours scatter across the plot — per-passage refinement has substituted in chunks from elsewhere in the corpus that match each individual question independently but no longer form a coherent set. **Right (HCPC-v2):** per-query clusters return, because the selective refinement policy declines to refine when doing so would shatter the original cluster.

13.4 Interpretation

If P1–P3 hold, the paradox has a mechanistic correlate that goes beyond "the NLI detector fires more on fragmented contexts." Specifically, fragmentation shifts the generator from evidence-following to prior-following inside a span of layers that prior work has identified as the site of context integration. This reframes HCPC-v2's selective refinement as a *load-management* policy: it avoids driving the model into a regime where its attention is no longer anchored on the retrieval set.

We flag one important caveat. Attention is not a faithful explanation of model behavior in general [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019]. The probe here is used in a narrow, paired-difference setting where both conditions share the same question, model, and decoding parameters; we report per-layer deltas rather than raw values to avoid over-interpreting absolute attention weights. The mechanistic evidence complements, rather than replaces, the behavioural results in §7.

14 Adversarial Coherence Failures

The naturalistic experiments in §7 demonstrate the coherence paradox *in distribution*, but leave open the question of whether our coherence signals can isolate *specific* fragmentation patterns. This section reports an adversarial probe with hand-authored cases that target three failure modes individually.

14.1 Case design

We construct 40 cases organized into four subsets of 10 (data files in `data/adversarial/`; loader at `src/adversarial_cases.py`). Each case is a tuple `(case_id, category, question, passages, expected_failure_mode, notes)`. The categories are:

DISJOINT passages that are individually relevant to the question but drawn from disjoint topical neighborhoods. Example: "*What is the primary cause of myocardial infarction?*" with passages alternating between cardiology, ICU triage, and sports-medicine recovery — every passage matches a query keyword but none cohere.

CONTRADICT relevant passages containing contradictory claims about the queried entity (e.g., one passage says aspirin reduces stroke risk, another says it raises bleeding risk in the same cohort). Faithfulness here cannot be measured against a single ground-truth answer; the targeted signal is the contradiction itself.

DRIFT passages where the topic gradually drifts from a near-miss neighbor to the true target ("capital of Australia" passages drifting Sydney → Canberra → ACT museums). The model must locate the answer in only one of the passages.

CONTROL matched coherent controls: passages that share the same surface topic and length distribution but do not exhibit any of the three pathologies. These calibrate signal magnitudes against a non-adversarial baseline.

The cases are released as part of ContextCoherenceBench (§15) so other researchers can probe their own retrievers.

14.2 Detection task

For each case we compute eight retrieval-time signals:

1. `mean_query_chunk_sim` (negated)
2. `ccs` (negated)
3. `mean_jaccard` (negated)
4. `embedding_variance`
5. `retrieval_entropy`
6. `sim_spread`
7. `nli_pairwise_mean`
8. `nli_pairwise_max`
9. `nli_pairwise_frac_hi`

Signals (1)–(3) are negated so that, for every signal, "higher = more adversarial" under the predicted direction. We then report binary detection AUC (sklearn `roc_auc_score`) for each (signal, adversarial-category-vs-control) pair.

The `nli_pairwise_*` family is new in this paper: `compute_nli_pairwise` (in `src/coherence_metrics.py`) runs the same NLI cross-encoder used for faithfulness scoring on every pair of retrieved passages and reports the mean / max contradiction score and the fraction of pairs above a contradiction threshold. This gives the CONTRADICT category a signal that pure embedding-space metrics cannot capture by construction.

14.3 Preregistered analysis plan

As with the mechanistic probe, we publish the analysis plan before the numbers so that the scoring cannot drift to fit the signal panel. The final reporting unit is a single table of detection AUCs: three rows (one per adversarial category: DISJOINT, CONTRADICT, DRIFT) and eight columns (one per coherence signal listed above). Each cell is the AUC for separating the row's adversarial subset from the matched CONTROL subset using that column's signal; cells at AUC ≥ 0.80 are marked in bold. We also report 95% bootstrap confidence intervals over the 10 paired items per row so a reader can tell high-but-noisy cells apart from high-and-stable ones.

We commit in advance to two success conditions: (i) at least one coherence signal achieves AUC ≥ 0.80 on each of the three adversarial categories, and (ii) the best signal for

CONTRADICT comes from the `nli_pairwise_*` family rather than from any embedding-space signal, which is the dissociation the category was designed to produce. A uniform AUC table — whether uniformly high, uniformly low, or uniformly middling — counts as a failed dissociation and will be reported as written.

14.4 What we expect to learn

The categories were designed to dissociate signals:

- DISJOINT should be detected best by `ccs`, `mean_jaccard`, and `embedding_variance` — embedding-space coherence is exactly what fragments under topical disjointness.
- CONTRADICT should be detected by `nli_pairwise_*` and only weakly by embedding signals, because contradictory passages can be *embedded* close together (similar topic, opposite claim).
- DRIFT should be detected by `sim_spread` and `retrieval_entropy` — drift produces a long tail of decreasing similarity rather than a homogeneous low-similarity set.

If this dissociation holds empirically, the practical implication is that no single coherence signal is sufficient: a deployment monitor wanting to catch all three failure modes needs at least one embedding-space signal *and* an NLI-pair signal. We therefore recommend treating the signals as a panel rather than a scalar.

Honest negative case. If the AUC table shows uniformly high or uniformly low values across categories and signals, the probe has failed to distinguish failure modes — either the cases are too easy / too hard, or our signal panel is collinear in practice. We will report this outcome as written if it occurs; the probe is informative under both results.

15 ContextCoherenceBench

To enable replication and follow-on work we release **ContextCoherenceBench**, a public benchmark targeting the three empirical phenomena reported in this paper. The release is bundled by `scripts/package_benchmark.py` and shipped to HuggingFace Datasets and a companion GitHub repository (URLs withheld for double-blind review; both are tagged `bench-v1` at the same git SHA).

15.1 Tasks

`adversarial_coherence` 40 hand-authored cases (10 DISJOINT / 10 CONTRADICT / 10 DRIFT / 10 CONTROL). Target metric: per-signal detection AUC against the matched control set, scored under the preregistered plan in §14.3. Designed to dissociate which coherence signal catches which fragmentation pattern.

`coherence_paradox` the multi-dataset query set (§8) with per-query measurements for five conditions (baseline, hcpc_v1, hcpc_v2, crag, selfrag). Target metrics: `paradox_drop` and `v2_recovery`, under the reporting plan in §8.2. Lets researchers test whether their own retrievers / generators reproduce the paradox.

`human_faithfulness` 500 double-annotated items with binary faithfulness, 1–5 correctness, 1–5 helpfulness, and Fleiss' κ. Target metric: NLI $\leftrightarrow$ human Spearman correlation, under the reporting plan in §11.2. Lets follow-on work calibrate new faithfulness scorers against the same human ratings.

15.2 Layout

The release directory mirrors the HuggingFace dataset spec:

```
release/context_coherence_bench_v1/
  README.md                  (HF dataset card)
  CITATION.bib  LICENSE
  metadata.json              (provenance + sha256 per file)
  adversarial/
    {disjoint,contradict,drift,control}.jsonl
  coherence_paradox/
    per_query.csv
    summary.csv
  human_faithfulness/
    aggregated.csv
    manifest.csv
```

The hand-authored adversarial cases are released under **CC-BY-4.0**. Per-dataset content inherits the upstream license recorded in `metadata.json`; the human-annotation ratings are **CC-BY-4.0** with annotator IDs redacted to opaque `A<n>` strings.

15.3 Intended use

We see three productive uses for the benchmark:

1. **Coherence-signal research.** The `adversarial_coherence` subset isolates failure modes cleanly enough that a new signal proposal can be evaluated on category-stratified AUC rather than only aggregate detection.

2. **Selective-refinement comparison.** The `coherence_paradox` subset already reports five conditions on the same queries; a new retrieval method (e.g., a sixth condition) can be added without re-running ours.

3. **Faithfulness-scorer calibration.** The `human_faithfulness` subset gives any new automatic-faithfulness model a fixed reference set to validate against, including the paradox-survival check.

We *do not* intend the benchmark as a leaderboard. The metric the paper centers on (`paradox_drop`) is a diagnostic, not a performance score; ranking systems by it would invert the intended use. The release ships scripts but no scoreboard.

15.4 Reproduction commands

The companion repository ships entry points matching each subset:

```
python3 experiments/run_adversarial_coherence.py
python3 experiments/run_multidataset_validation.py
python3 experiments/run_headtohead_comparison.py
python3 experiments/run_mechanistic_analysis.py
python3 experiments/prepare_prolific_study.py
python3 experiments/analyze_prolific_results.py
python3 scripts/package_benchmark.py
```

Each script writes a self-contained results directory under `results/` that the packager copies into the release tree. Re-running any single experiment refreshes only its own subset, so partial reruns remain coherent.

16 Conclusion

The central finding of this paper is uncomfortable, because it contradicts a premise most RAG engineering quietly depends on: that improving retrieval quality will improve answer faithfulness. Across 6,050+ queries spanning two datasets, two model families, and four retrieval configurations, we show the premise is unreliable — and in some regimes inverted. Naive post-retrieval refinement improves local matching scores while cutting faithfulness by up to 15.9 percentage points; domain-mismatched retrieval can make a RAG system *less* faithful than the same model with no retrieval at all.

Once we stop treating retrieval as a per-passage problem, the pattern resolves. Faithfulness depends on whether the retrieved passages form a mutually supportive evidence set, not merely on whether each passage looks individually relevant. HCPC-v2 helps not because it is clever, but because it is conservative: it intervenes only when two independent signals both flag a fragmentation risk, protects the highest-ranked passage, and restores contiguity whenever it does refine. CCS is the same idea compressed into a number a practitioner can read off at retrieval time — before any generation has happened, and before any downstream harm.

Three findings have immediate practical value. First, chunk size is the dominant design parameter for faithfulness among the factors tested (90.1% of between-factor variance on SQuAD), outweighing prompt engineering by a factor of roughly 450×. Second, post-retrieval refinement should be applied selectively based on context coherence, not as a default post-processing step. Third, coherence should be validated before deploying RAG in any specialized domain where the encoder is not matched to the corpus.

What we did not settle, and where to push next. The paradox is well-characterized at the scale we ran, but load-bearing questions remain open. Scaling is one: does a 70B+ generator, with more parametric capacity to fall back on, amplify the failure mode or absorb it? Encoder alignment is another: fitting a domain-matched encoder (BioLinkBERT, PubMedBERT) to the PubMedQA corpus should test whether the coherence account predicts the right repair. Multi-hop and conversational settings are the cases where we expect coherence to matter *more*, not less, because the generator must stitch evidence across reasoning steps; if the paradox does not appear there, the account needs rethinking. Finally, CCS is used here as a diagnostic; turning it into an active objective — a retriever that optimizes set-level coherence rather than per-passage relevance — would move coherence from something we measure to something we directly retrieve for. We see that last direction as the most likely to produce a qualitatively different class of RAG systems, and the most in need of careful evaluation before anyone ships it.

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A Hyperparameters

Table 13 lists all hyperparameters used across experimental configurations.

Table 13: Complete hyperparameter specification.

Component	Parameter	Value	Notes
Embedding			
	Model	all-MiniLM-L6-v2	82M params, 384-dim
	Normalization	L2	cosine similarity
Chunking (fixed)			
	Chunk sizes	{256, 512, 1024}	tokens
	Overlap	10%	26, 51, 102 tokens
	Separators	`["\n\n", "\n", ".  ", " ", ""]`	recursive
Retrieval			
	Top-k	{3, 5}	cosine similarity
	Vector store	ChromaDB v0.4	persistent
Generation			
	Models	Mistral-7B, Llama-3-8B	via Ollama v0.1.29
	Temperature	0.1	low stochasticity
Cross-encoder			
	Model	ms-marco-MiniLM-L-6-v2	66M params
	Output	raw logits	not probabilities
NLI scorer			
	Model	nli-deberta-v3-base	435M params
	Threshold	0.50	hallucination cutoff
HCPC-v2			
	τ_{sim}	0.45	cosine threshold
	τ_{ce}	-0.20	CE logit threshold
	N_{prot}	2	rank protection
	N_{max}	2	max refine per query
	Sub-chunk size	256 tokens	for splitting
	Sub-overlap	32 tokens	
	Merge max	4096 chars	≈1024 tokens
Reranker			
	Fetch-k	7	over-fetch depth
	Return-k	3	final context size

B Chunk size ablation and variance analysis

Tables 14 and 15 report the full 12-configuration ablation over chunk size, top-k, and prompt strategy. All experiments use Mistral-7B with $n = 30$ queries per configuration.

Chunk-size summary. Mean faithfulness by chunk size on SQuAD: 256 = 0.7018, 512 = 0.7801, 1024 = 0.8105. Mean hallucination: 256 = 9.17%, 512 = 2.5%, 1024 = 2.5%.

Table 14: Full chunk size ablation on SQuAD (Mistral-7B, $n = 30$).

Chunk	k	Prompt	Faithfulness	Halluc.
256	3	strict	0.6732	10.0%
256	3	cot	0.6835	6.7%
256	5	strict	0.7254	10.0%
256	5	cot	0.7253	10.0%
512	3	strict	0.7827	0.0%
512	3	cot	0.7781	0.0%
512	5	strict	0.7861	3.3%
512	5	cot	0.7737	6.7%
1024	3	strict	0.8001	3.3%
1024	3	cot	0.7891	0.0%
1024	5	strict	0.8274	3.3%
1024	5	cot	0.8254	3.3%

Table 15: Full chunk size ablation on PubMedQA (Mistral-7B, $n = 30$).

Chunk	k	Prompt	Faithfulness	Halluc.
256	3	strict	0.5591	30.0%
256	3	cot	0.5802	20.0%
256	5	strict	0.5585	30.0%
256	5	cot	0.5636	26.7%
512	3	strict	0.6028	16.7%
512	3	cot	0.6196	13.3%
512	5	strict	0.5844	20.0%
512	5	cot	0.5906	13.3%
1024	3	strict	0.5963	20.0%
1024	3	cot	0.6167	10.0%
1024	5	strict	0.5849	16.7%
1024	5	cot	0.5881	13.3%

On PubMedQA: faithfulness 256 = 0.5654, 512 = 0.5994, 1024 = 0.5965; hallucination 256 = 26.67%, 512 = 15.83%, 1024 = 15.0%. The 256 → 1024 improvement is large on SQuAD (10.87 pp faithfulness, 6.67 pp hallucination reduction) and moderate on PubMedQA (3.11 pp faithfulness, 11.67 pp hallucination reduction).

B.1 ANOVA results

Table 16 reports one-way ANOVA results for each factor.

Chunk size is the only factor reaching significance on both datasets. On SQuAD, $F = 19.72$ with $p < 0.001$; on PubMedQA, $F = 3.66$ with $p = 0.027$. Prompt strategy is non-significant on both datasets ($p = 0.830$ on SQuAD, $p = 0.290$ on PubMedQA), and

Table 16: One-way ANOVA for faithfulness by factor.

Dataset	Factor	F	p	η^2	Sig.
SQuAD	Chunk size	19.720	<0.001	0.100	Yes
SQuAD	Prompt strategy	0.046	0.830	<0.001	No
SQuAD	Top-k	2.918	0.089	0.012	No
PubMedQA	Chunk size	3.664	0.027	0.020	Yes
PubMedQA	Prompt strategy	1.122	0.290	0.005	No
PubMedQA	Top-k	2.323	0.128	0.010	No

the effect size is negligible ($\eta^2 < 0.005$).

B.2 Cohen's d effect sizes

Table 17 reports pairwise Cohen's d between factor levels.

Table 17: Cohen's d effect sizes for faithfulness.

Comparison	SQuAD	PubMedQA
Chunk size		
256 vs 512	0.82	0.31
256 vs 1024	1.41	0.29
512 vs 1024	0.55	0.03
Prompt strategy		
strict vs CoT	0.02	0.11
Top-k		
$k = 3$ vs $k = 5$	0.18	0.16

The 256-to-1024 chunk size effect on SQuAD ($d = 1.41$) is large by conventional standards ($d > 0.8$). The 512-to-1024 effect ($d = 0.55$) is medium. On PubMedQA, all chunk size effects are small ($d < 0.35$), reflecting the domain mismatch that limits the benefit of larger chunks. Prompt strategy effects are negligible on both datasets ($d \leq 0.11$).

B.3 Variance decomposition by model

Table 18 partitions the variance in faithfulness among the three configuration factors, computed separately for each model-dataset combination. This uses the multi-model data from the extended experiments (Llama-3-8B and Mistral-7B, $n = 100$ per condition).

Table 18: Variance decomposition: proportion of faithfulness variance explained by each factor, by model and dataset.

Model	Dataset	Chunk size	Top-k	Prompt
Overall				
—	SQuAD	90.1%	9.8%	0.2%
—	PubMedQA	61.2%	26.1%	12.7%
By model				
Mistral-7B	SQuAD	72.4%	9.0%	18.6%
Llama-3-8B	SQuAD	53.3%	25.2%	21.5%
Mistral-7B	PubMedQA	1.5%	97.9%	0.6%
Llama-3-8B	PubMedQA	95.2%	0.4%	4.4%

The model-level breakdown reveals substantial heterogeneity. On SQuAD, chunk size dominates for both models (72.4% for Mistral, 53.3% for Llama-3), consistent with the overall 90.1% figure. On PubMedQA, the models diverge sharply: Mistral's faithfulness is governed by top-k (97.9%), while Llama-3's is governed by chunk size (95.2%). This

suggests that optimal RAG configurations are model-specific and should not be assumed to transfer between architectures.

Why chunk size matters. Chunk size affects faithfulness through three reinforcing mechanisms. First, larger chunks contain complete sentences and paragraphs; a 256-token chunk may truncate mid-thought, producing lower NLI entailment scores even for qualitatively correct answers. Second, with $k = 3$ and chunk = 1024, the generator receives approximately 3,000 tokens of context versus 750 at chunk = 256; more context provides redundant evidence that reduces hallucination risk. Third, larger chunks preserve the logical flow of source documents—topic sentences, supporting details, and transitions appear together, maintaining the coherence that smaller chunks fragment.

The 256-to-1024 effect on SQuAD ($d = 1.41$) reflects all three mechanisms acting together. The 512-to-1024 effect ($d = 0.55$) is smaller because 512-token chunks already capture most complete sentences and moderate redundancy.

Why prompt strategy does not matter. The maximum faithfulness difference between strict and chain-of-thought prompts on SQuAD, holding chunk size and top-k constant, is 0.0121 (at chunk = 1024, $k = 3$: strict = 0.8001, CoT = 0.7891). Both prompts instruct the model to answer from the context. When the context is coherent and contains the answer, both produce faithful responses; when the context is fragmented or the answer is absent, neither can prevent hallucination. For RAG faithfulness, prompt engineering yields returns roughly 450× smaller than chunk-size selection ($\eta^2_{\text{chunk}}/\eta^2_{\text{prompt}} = 0.901/0.002$).

C Re-ranking by chunk size

Table 19 reports re-ranking results across chunk sizes (Llama-3-8B, $n = 100$ on SQuAD, $n = 50$ on PubMedQA).

Table 19: Re-ranking effect by chunk size. Δ columns show reranked minus baseline.

Dataset	Chunk	Base Faith.	Rerank Faith.	Δ Faith.	Δ Halluc.
SQuAD	256	0.7233	0.7265	+0.003	−3.0 pp
SQuAD	512	0.7523	0.7544	+0.002	−3.0 pp
SQuAD	1024	0.7739	0.7619	−0.012	0.0 pp
PubMedQA	256	0.6136	0.6455	+0.032	−10.0 pp
PubMedQA	512	0.5862	0.5957	+0.010	+6.0 pp
PubMedQA	1024	0.5853	0.5806	−0.005	−2.0 pp

Re-ranking interacts with chunk size in a non-obvious way. At small chunk sizes (256), re-ranking consistently helps: it improves faithfulness and reduces hallucination on both datasets, with the largest effect on PubMedQA (−10 pp hallucination). At large chunk sizes (1024), re-ranking slightly *hurts* faithfulness on both datasets. At intermediate sizes (512 on PubMedQA), re-ranking improves faithfulness but *increases* hallucination by 6 pp.

The pattern suggests that re-ranking is most useful when the initial retrieval is noisy (small chunks produce more candidates of variable quality) and least useful when chunks are already large enough to be self-contained. At chunk = 1024, the cross-encoder's preference for keyword-matching relevance can override the embedding model's broader semantic ranking, replacing a contextually appropriate passage with one that matches query terms more precisely but contributes less to overall coherence.

D Multi-model chunk size comparison

Table 20 reports faithfulness and hallucination rates for both models across chunk sizes (extended experiments, $n = 100$ per condition on SQuAD).

Table 20: Multi-model comparison by chunk size.

Model	Dataset	Chunk 256	Chunk 512	Chunk 1024
Faithfulness				
Llama-3	SQuAD	0.7184	0.7530	0.7591
Mistral	SQuAD	0.7248	0.7548	0.7676
Llama-3	PubMedQA	0.6151	0.5851	0.5868
Mistral	PubMedQA	0.5798	0.5797	0.5818
Hallucination rate				
Llama-3	SQuAD	4.5%	1.75%	1.0%
Mistral	SQuAD	8.25%	4.5%	3.75%
Llama-3	PubMedQA	16.0%	19.0%	18.5%
Mistral	PubMedQA	26.5%	28.5%	25.5%

Both models show the same chunk-size trend on SQuAD: larger chunks produce higher faithfulness and lower hallucination. On PubMedQA, the trend is weaker and non-monotonic for Llama-3 (faithfulness peaks at chunk = 256), consistent with the domain mismatch limiting the benefit of more context when that context is poorly matched.

E Threshold sweep details

The threshold sweep tested 168 fixed configurations (7 τ_{sim} values $\times$ 6 τ_{ce} values $\times$ 4 N_{prot} values) plus 4 adaptive configurations per dataset. Evaluation used the proxy score:

$$\text{proxy} = 0.4 \cdot \text{CCS} - 0.3 \cdot \text{var}(S) - 0.2 \cdot H(S) + 0.1 \cdot \bar{q} \tag{2}$$

where $\text{var}(S)$ is the embedding variance, $H(S)$ is the retrieval entropy, and $\bar{q}$ is the mean query-chunk similarity.

On SQuAD, the top-10 fixed configurations all achieve CCS = 1.0 with proxy score = 0.2587, compared to the default configuration's 0.2425. The optimal region is broad: configurations with $\tau_{\text{sim}} \geq 0.35$ and $N_{\text{prot}} = 0$ produce identical proxy scores because the AND gate rarely fires when the embedding model matches the corpus.

On PubMedQA, the top-10 configurations achieve proxy score = 0.2820 (default: 0.2496, $\Delta = +13\%$). These concentrate at $\tau_{\text{sim}} \in [0.50, 0.60]$ with $N_{\text{prot}} = 0$, where more chunks are eligible for refinement and the proxy rewards higher coherence after refinement.

Adaptive (percentile-based) thresholds underperform fixed thresholds on both datasets. The best adaptive configuration matches the default fixed configuration (proxy = 0.2425 on SQuAD, 0.2496 on PubMedQA). Fixed thresholds, once tuned to the dataset, are sufficient; adaptive adjustment adds complexity without benefit.

F Per-query failure analysis

We manually examined hallucinated responses from HCPC-v1 on SQuAD ($n = 10$). Three failure patterns emerged:

1. **Evidence truncation** (6/10): The refined sub-chunk did not contain the sentence needed to answer the question. The model generated a plausible but unsupported answer using its parametric memory.

2. **Context gap bridging** (3/10): The model inferred a connection between two non-adjacent sub-chunks that was not supported by either. The gap left by refinement was filled with fabricated reasoning.

3. **Over-completion** (1/10): The model extended its answer beyond what the context supported, appending details from parametric memory after correctly answering the initial question.

Under HCPC-v2, the only failure mode observed (3/3 hallucinated responses in extended experiments) was evidence truncation, and all three cases occurred on queries where the rank protection gate was active. This suggests that selective refinement occasionally prevents refinement of a passage that genuinely needs it, when the weak chunk happens to rank in the top 2.

G Latency measurements

Table 21: Mean generation latency (seconds, ± std dev).

Model	Dataset	Chunk = 256	Chunk = 512	Chunk = 1024
Llama-3	SQuAD	2.20 ± 1.16	3.23 ± 1.01	4.19 ± 1.59
Llama-3	PubMedQA	6.77 ± 43.3	5.87 ± 1.70	8.71 ± 2.78
Mistral	SQuAD	3.41 ± 1.78	4.56 ± 1.68	9.04 ± 91.3
Mistral	PubMedQA	6.37 ± 2.15	10.3 ± 3.28	13.0 ± 3.80

Latency scales roughly linearly with chunk size (more context tokens to process). The high standard deviations on some cells (e.g., Llama-3 PubMedQA chunk = 256: ±43.3s) reflect occasional outlier queries where the model produced unusually long responses.

The HCPC-v2 refinement step adds 2–3 seconds per query (cross-encoder scoring + sub-splitting). On SQuAD, where refinement triggers for 16.7% of queries, the amortized overhead is approximately 0.4 seconds per query. On PubMedQA, where 73.3% of queries trigger refinement, the overhead is approximately 1.8 seconds per query.

H Implementation details

The pipeline is implemented in Python 3.11. Key dependencies:

- `langchain` and `langchain-community` for text splitting and Ollama integration
- `sentence-transformers` for embedding and cross-encoder inference
- `chromadb` v0.4 for vector storage with cosine distance
- `torch` with MPS backend for Apple Silicon acceleration
- `scipy` for ANOVA and correlation tests
- `pandas` for result aggregation

The HCPC-v2 retriever (`src/hcpc_v2_retriever.py`, 550 lines) is the most complex component. It maintains per-query diagnostic logs that record: whether refinement was triggered, the number of chunks before and after refinement, the best cosine and cross-encoder scores before and after, the CCS, approximate token count, threshold mode, and effective threshold values.

All experiments use the same 30 (or 100, depending on phase) queries per condition, sampled randomly from the validation split at the start of each experiment and held fixed across all conditions within that experiment. This ensures that condition comparisons are not confounded by query difficulty variation.

I Real-world case studies

To make the refinement paradox concrete beyond the aggregate metrics, we present a small case-study set: queries on which the same generator gives a faithful answer with raw retrieval, hallucinates after per-passage refinement, and recovers under the coherence-gated policy. The cases are auto-selected from the per-query logs by the script `experiments/build_qualitative_example.py` as the queries with the largest baseline→HCPC-v1 faithfulness gap.

Case 1. *Dataset:* squad *Model:* mistral
Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*
Baseline ($f = 0.942$): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.
HCPC-v1 ($f = 0.478$): San Francisco Bay Area
HCPC-v2 ($f = 0.942$, recovered): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Case 2. *Dataset:* squad *Model:* mistral
Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*
Baseline ($f = 0.942$): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.
HCPC-v1 ($f = 0.478$): San Francisco Bay Area
HCPC-v2 ($f = 0.942$, recovered): The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Case 3. *Dataset:* squad *Model:* mistral
Question. What year did the Denver Broncos secure a Super Bowl title for the third time?
Ground truth. *2015*
Baseline ($f = 0.795$): The provided context does not specify the exact year when the Denver Broncos secured a Super Bowl title for the third time. However, it is implied that this event happened in Super Bowl 50, which was held in 2016.
HCPC-v1 ($f = 0.485$): I cannot find this information in the provided context. The context only mentions that the Denver Broncos won the AFC Championship Game and secured home-field advantage throughout the AFC playoffs, but it does not speci\{}dots
HCPC-v2 ($f = 0.795$, recovered): The provided context does not specify the exact year when the Denver Broncos secured a Super Bowl title for the third time. However, it is implied that this event happened in Super Bowl 50, which was held in 2016.

Case 4. *Dataset:* squad *Model:* llama-3.3-70b
Question. If Roman numerals were used, what would Super Bowl 50 have been called?
Ground truth. *Super Bowl L*
Baseline ($f = 0.502$): If Roman numerals were used, Super Bowl 50 would have been called Super Bowl L.
HCPC-v1 ($f = 0.491$): Super Bowl L.
HCPC-v2 ($f = 0.502$, recovered): If Roman numerals were used, Super Bowl 50 would have been called Super Bowl L.

Figure 17: **Real-world case studies of the refinement paradox.** Five queries where the same question is answered correctly with raw retrieval, incorrectly after per-passage refinement, and correctly again under coherence-gated refinement. Faithfulness f is the NLI entailment score (higher is better; ≥ 0.5 is the conventional faithful threshold). All cases satisfy baseline = faithful, HCPC-v1 = hallucinated, HCPC-v2 = faithful.